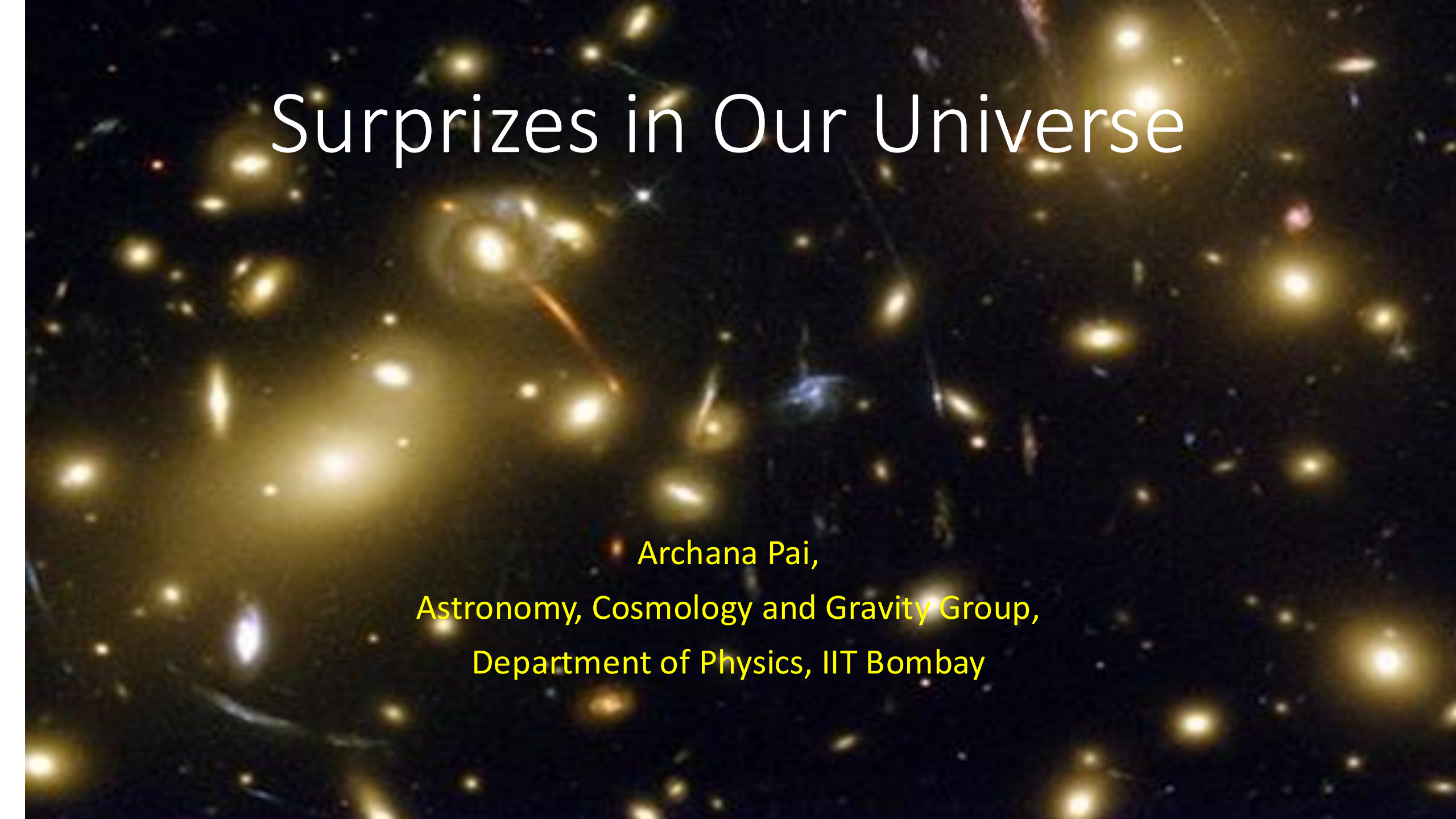


Surprises in Our Universe

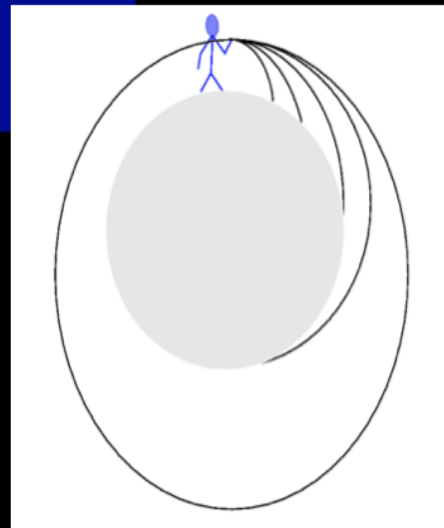
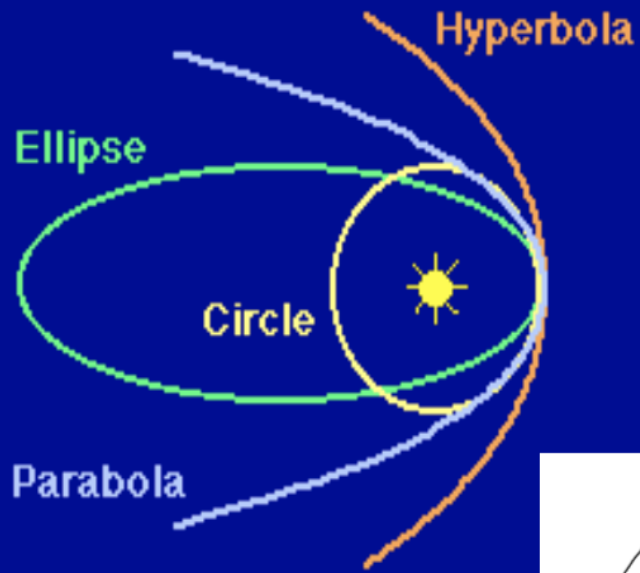
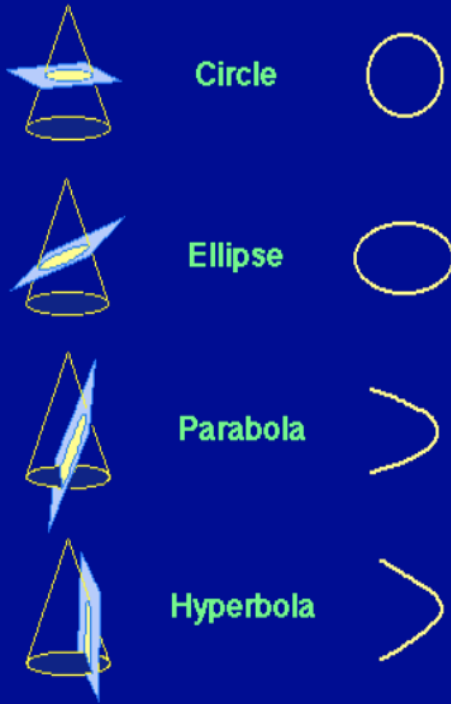


Archana Pai,
Astronomy, Cosmology and Gravity Group,
Department of Physics, IIT Bombay

Planetary Orbits Explained except Mercury

Law of Gravitation+Second Law of Newtonian mechanics
== Gives orbits

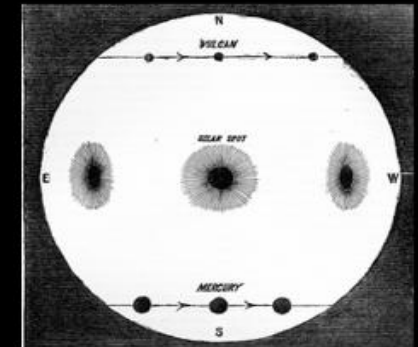
Conic Sections



French Mathematician, La Verrier
(1859)

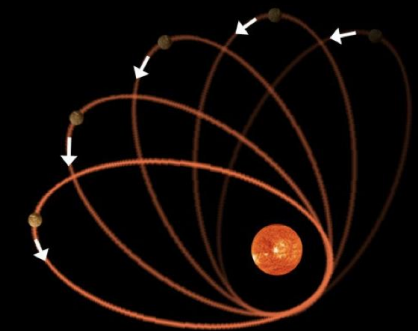


Extensive Study
of Mercury's
motion during
the transit
=>Anomalous
behaviour of
Mercury's orbit

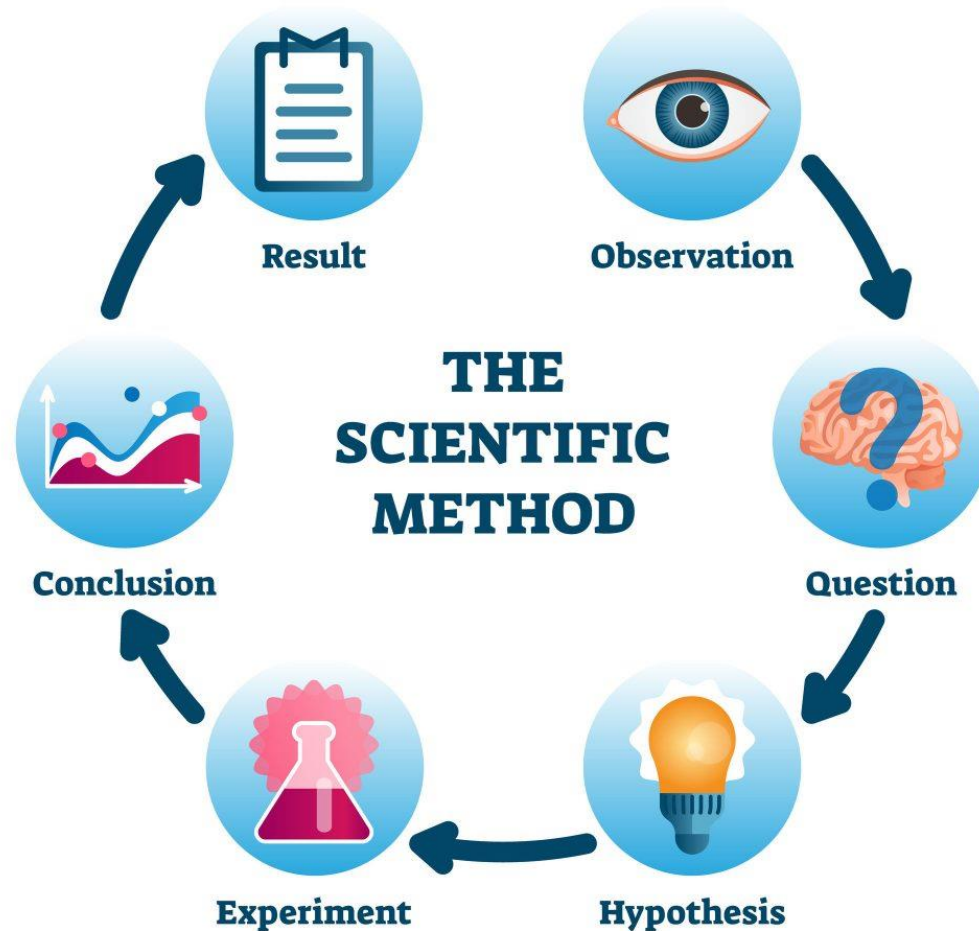


Observed shift:
574" per century
Effect from planets:
531" per century

Unaccounted shift:
43" per century



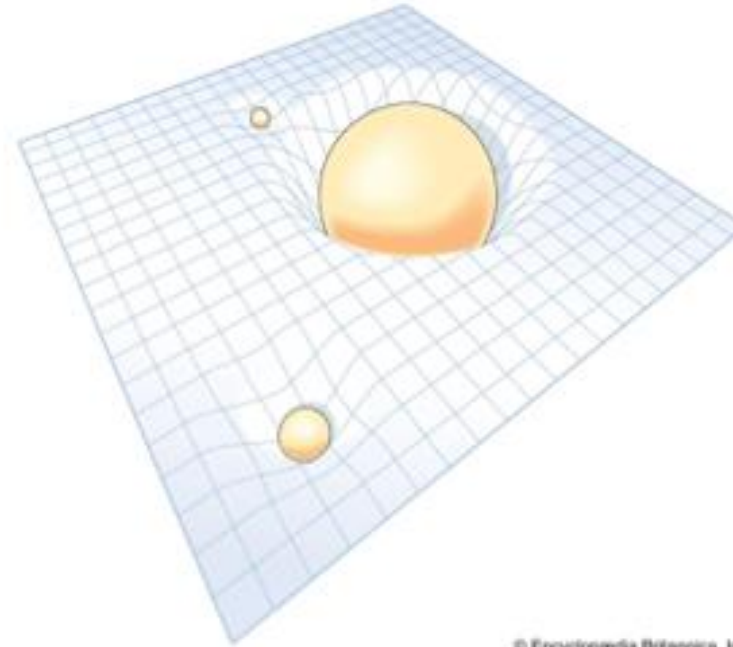
Scientific methodology:



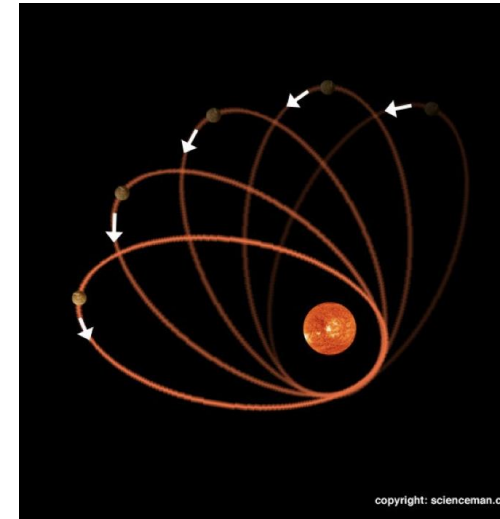
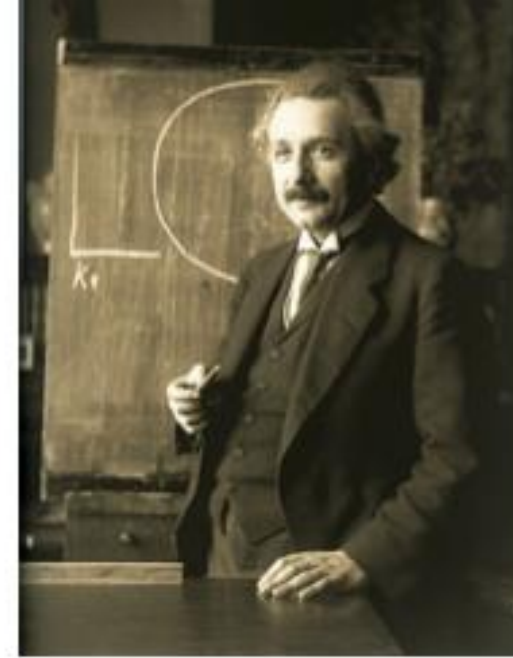
Does the motion of Mercury follow Newtonian Dynamics?

1915: Einstein's General Theory of Relativity

- **Strong Equivalence principle:** Freely falling frames are local inertial frames where laws of physics same.
- **Einstein's equations:** Motion under gravity provides motion of object in curved geometry.
- Prescription of curved geometry is given by the Riemannian geometry.
- **Matter defines geometry** and the geometry decides the trajectory.
- Under **weak gravity and small velocities** Newton's Law of gravitation.



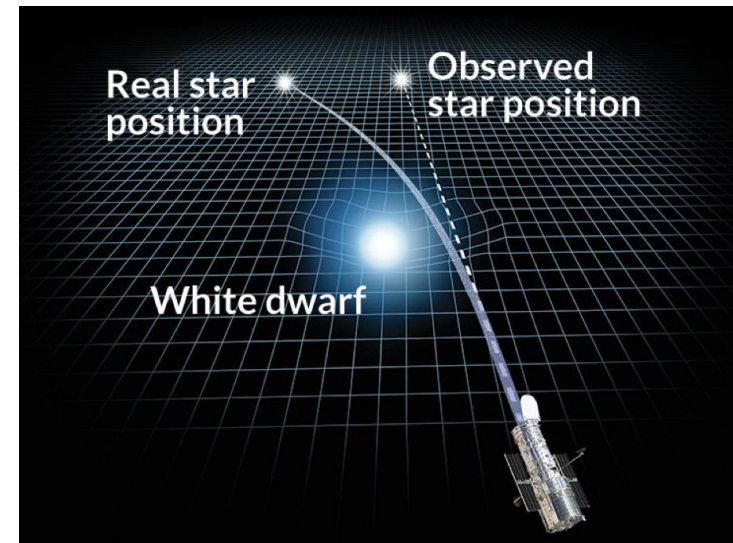
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Predictions of Einstein's General Relativity

- Under weak gravity and small velocities, the GTR gives Newton's Law of gravitation – Hence explains planetary motion
- New Predictions
 - Gravitational time dilation : Used in Global Positioning System
 - Bending of light : gravitational lensing
 - Black holes and compact objects
 - Expansion of the Universe
- New observations, new surprises



Eddington's Solar experiment on May 29th, 1919

LIGHTS ALL ASKEW IN THE HEAVENS

Men of Science More or Less
Agog Over Results of Eclipse
Observations.

EINSTEIN THEORY TRIUMPHS

Stars Not Where They Seemed
or Were Calculated to be,
but Nobody Need Worry.

A BOOK FOR 12 WISE MEN

No More in All the World Could
Comprehend It, Said Einstein When
His Daring Publishers Accepted It.

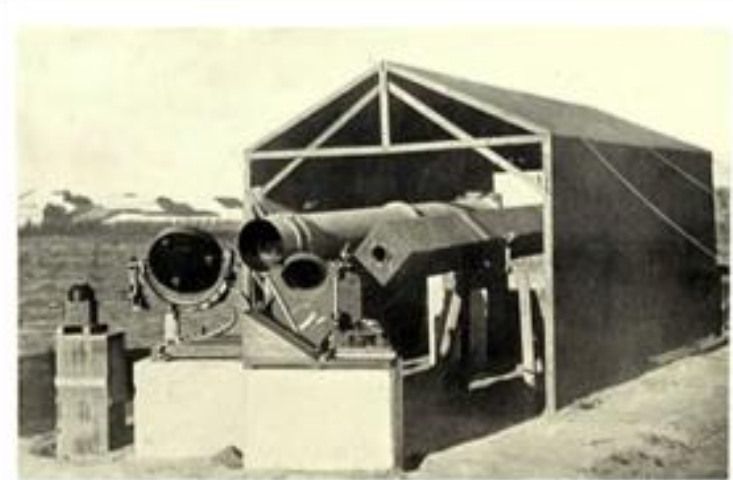
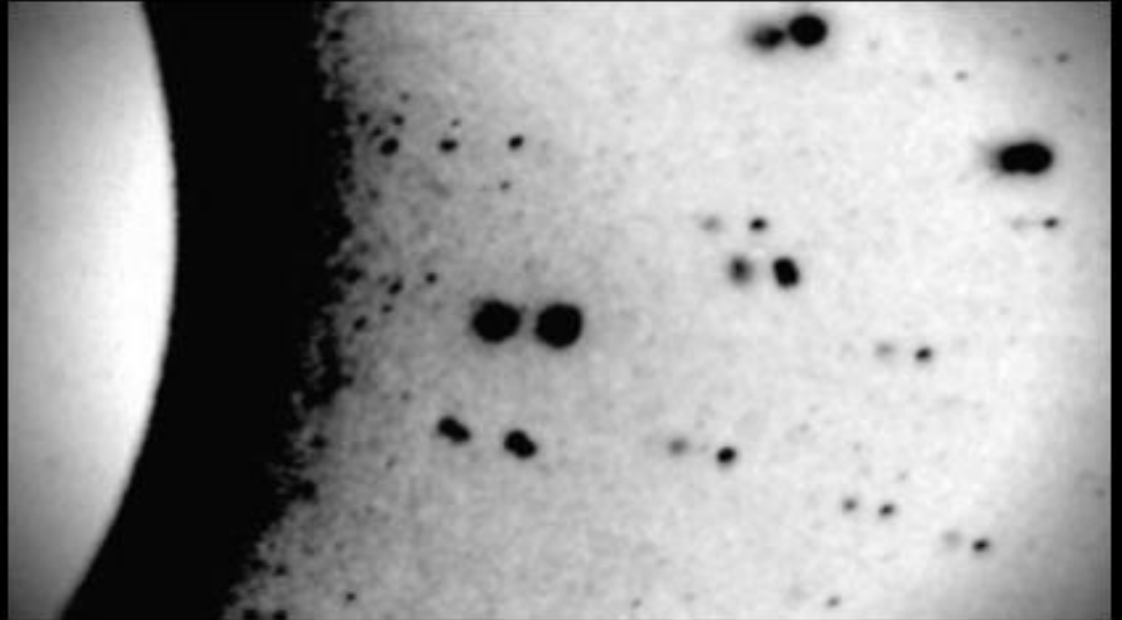
The New York Times of November 10,
1919, reported on Einstein's confirmed
prediction



Arthur Stanley Eddington, Plumian
Professor of Astronomy

[https://
www.wikiwand.
com/en/
Eddington_exp
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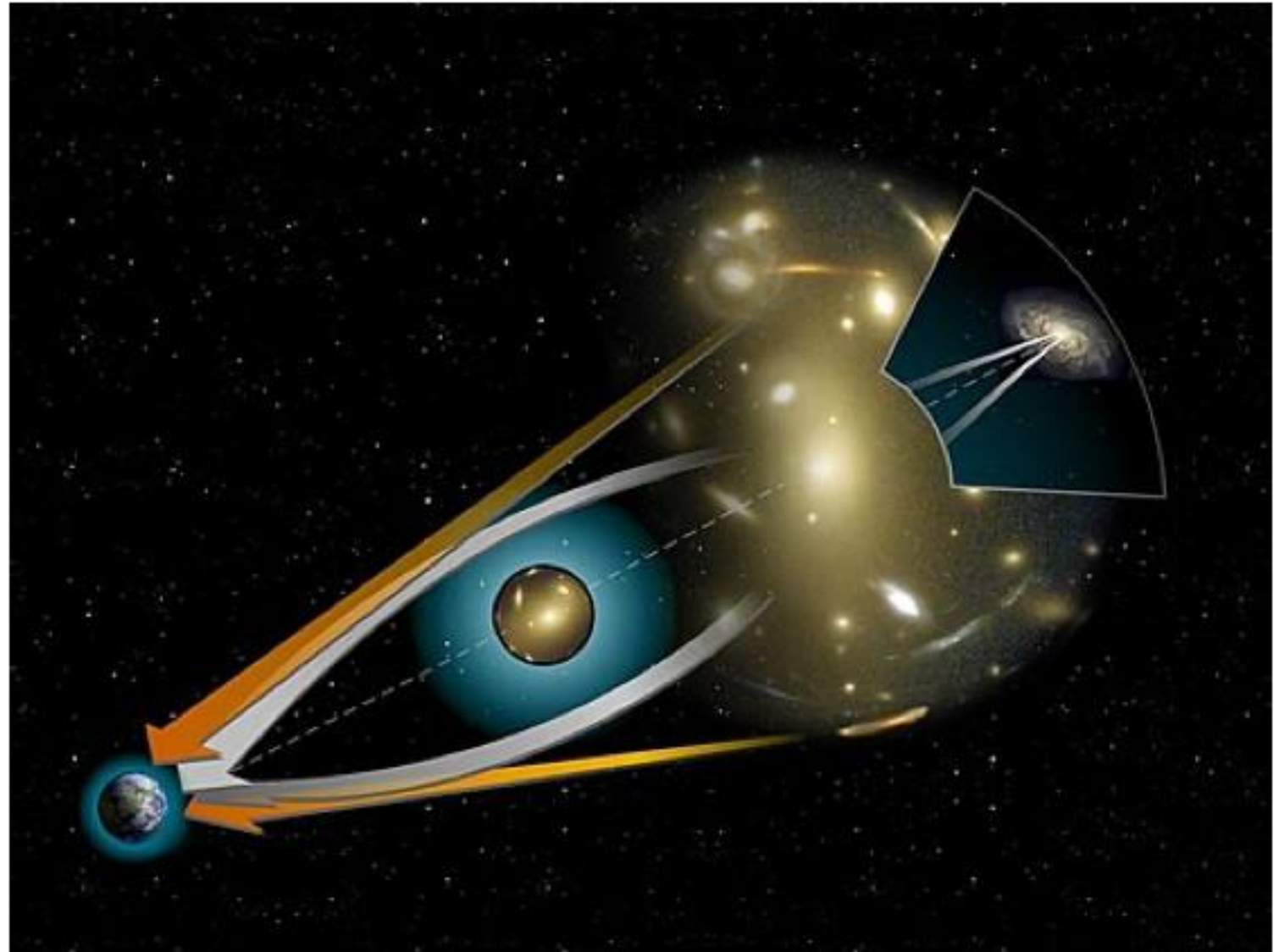
Watch — the
movie Einstein
and Eddington



Eclipse instruments at Sobral

Gravitational Lensing (1936 by Einstein)

- 1936: Einstein predicted gravitational lensing with a note that 'Of course, there is no hope of observing this phenomenon directly.'
- 1937: Zwicky predicted that galaxies can act as lenses.



Lensing discovery

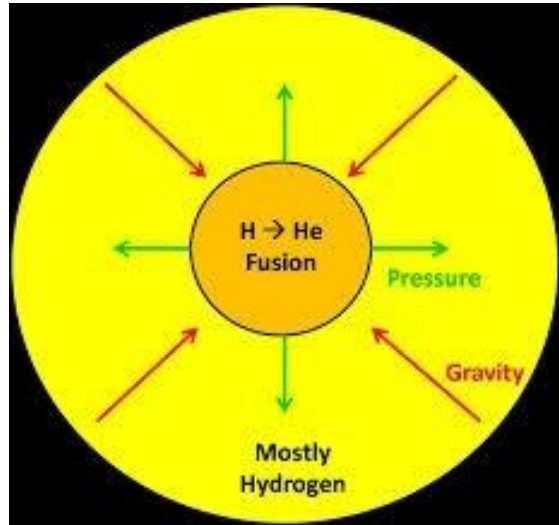
- 1979: Discovery of "Twin QSO" (SBS 0957+561) — Walsh, Carswell and Weyman observed two Quasars in close proximity. Two images of the same object lensed by the intervening galaxy.
- Separation of 6 arcsec with similar apparent magnitude of 16.7 and 16.5
- Lensing galaxy is G1, a giant elliptical galaxy within a cluster of galaxies
- Observed by 2.1 telescope at Kitt-Peak National Observatory in Arizona.





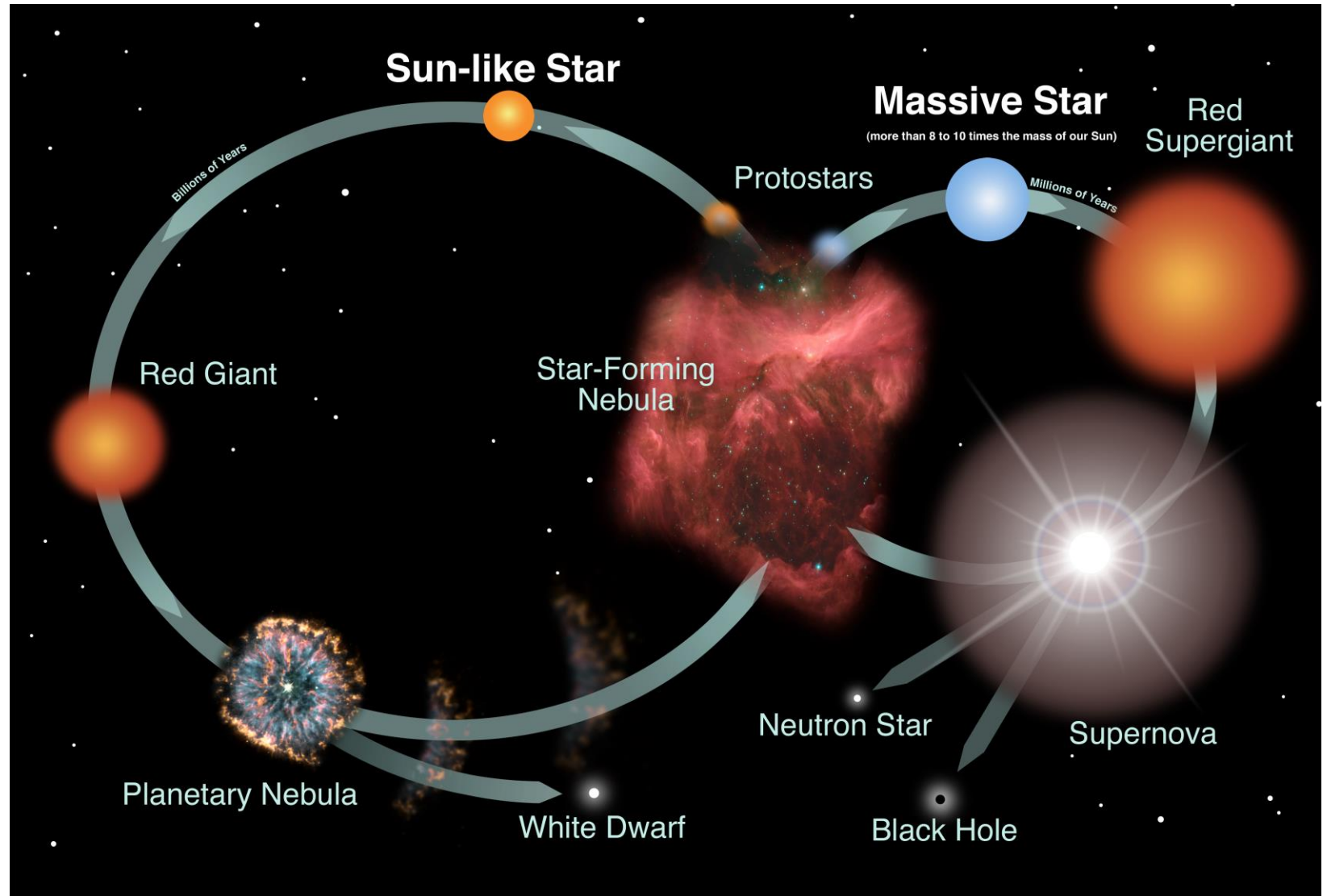
Compact Objects

Life cycle of massive stars



The Sun

Stars with mass $> 10 M_{\text{sun}}$
ends as either NS/BH



Compact objects

- Compactness parameter measures the strength of the gravity

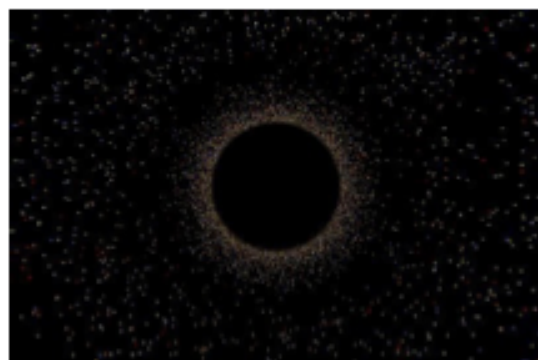


Sun

Mass ~ $1e30$ kg
Radius ~ 700,000 km
 $\alpha \sim 0.000004$

$$\alpha = \frac{2GM}{Rc^2}$$

M : Mass of the system
R: size of the system



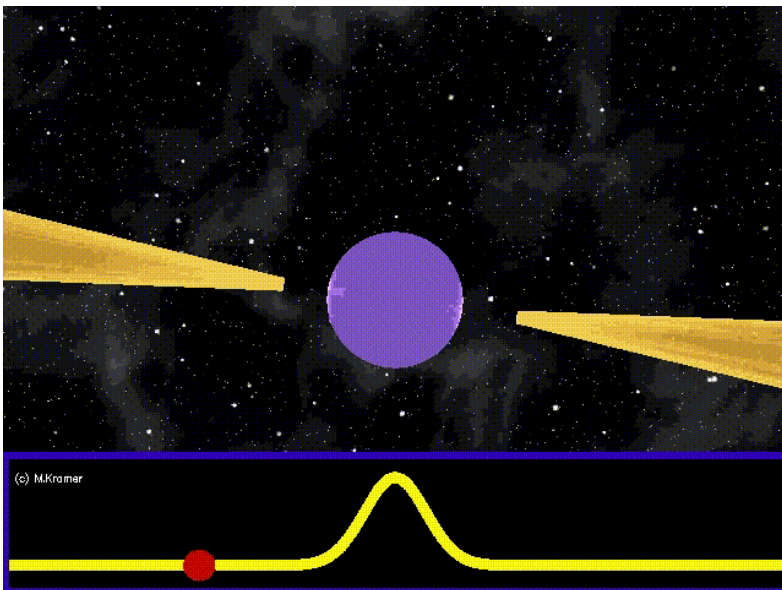
Black Hole (BH)

Mass ~ 3.0 M_{sun}
Radius ~ 9.0 km
 $\alpha \sim 1$

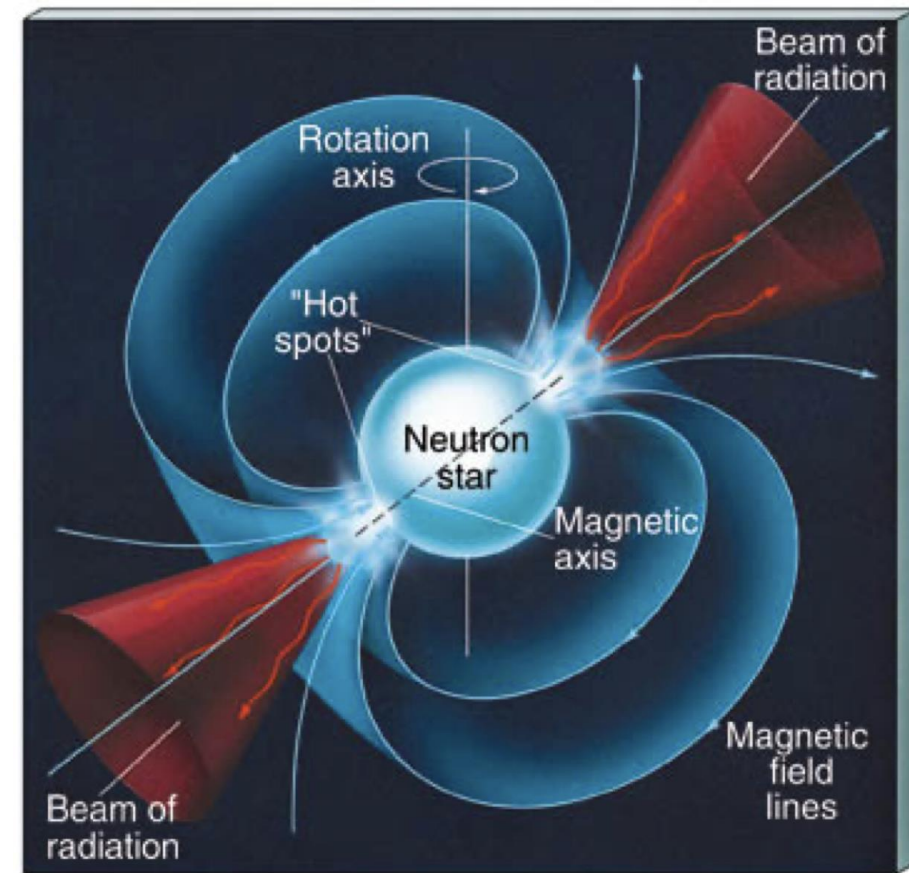
- In compact astrophysical systems, we need to account for GR

Pulsars: Nature's accurate clock

- End product of stellar evolution
- Rotating Neutron Star with neutrons and protons
- Radius $\sim 10\text{km}$, Mass $\sim 1.4\text{ Msun}$
- Magnetic field : $10^8\text{ G} - 10^{12}\text{ G}$
- Observed in radio band due to synchrotron emission
- Cosmic light houses



Joselin Bell, 1967



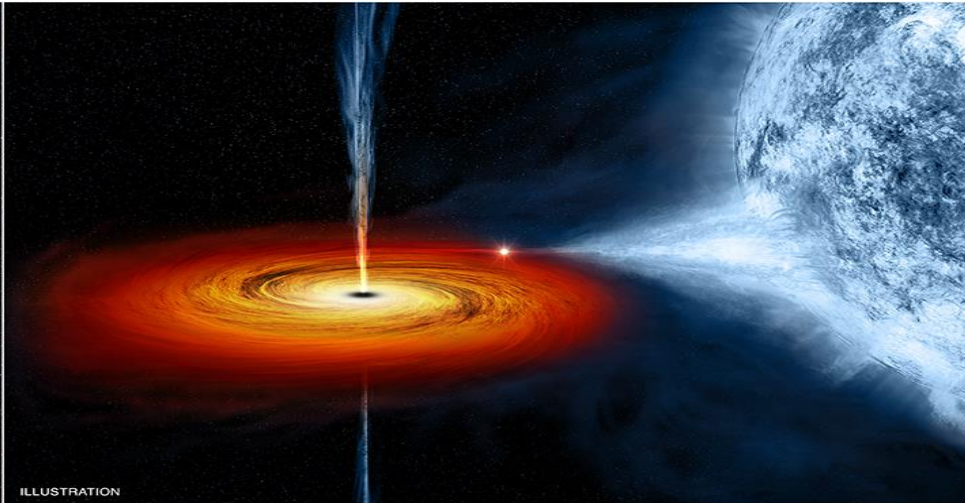
Antony Hewish (Nobel Prize in 1974 for the Pulsar discovery)

Open Questions:

Composition of Neutron Stars, StarQuakes
On the star, Magnetar??

Black Holes: Most compact Objects

- Extremely compact object with extreme gravity
- No particle or even light can escape from any black hole
- Theoretical prediction: Schwarzschild in 1916 , Was believed to be a mathematical construct, Kerr Metric (1963)
- Cyg-X1: First BH evidence in X-ray astronomy in 1964



Cygnus X-1

Stellar Mass Black Holes

$$(M < 100M_{\odot})$$

Most are formed by **direct collapse** of stellar cores

Inspiral, Merger and Ringdown of stellar mass BH binaries lie in the LIGO band

Intermediate Mass Black Holes

$$(M \in [10^2, 10^5]M_{\odot})$$

Stars with He core above $135M_{\odot}$ directly collapse into IMBH (**Population III**)

Hierarchical mergers in dense stellar environments

Merger and Post-Merger dominated signal

Supermassive Black Holes

$$(M > 10^5M_{\odot})$$

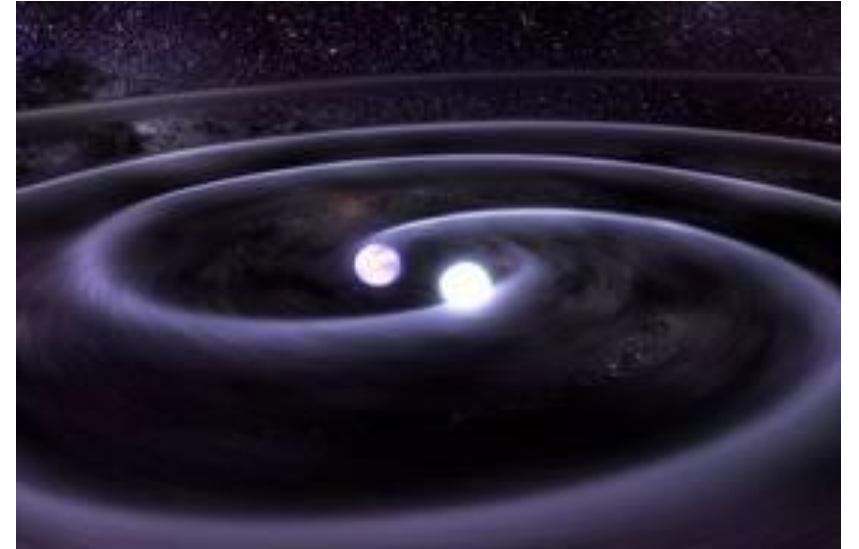
Hierarchical mergers of IMBHs leading to SMBHs

Collapse of **metal free gas clouds**

Growth of massive BH seeds via accretion

Observational avenues of Stellar mass BH

- **Observed in accreting systems (X-ray)**
 - Luminosity: $10^4 - 10^9$ Luminosity of Sun
 - Gravitational pull is higher than the radiation push
 - Stellar mass BHs e.g. Cygnus X- 1, 15 M_{sun} BH, highly spinning BH
- **Observed in binary mergers (GW)**
 - BHs in compact binary orbit
 - Emits gravitational waves.
 - BHs of various sizes are observed by LIGO detectors
 - First confident GW detection: GW150914
 - Luminosity of GW150914: 10^{56} ergs/sec. (10^{26} Luminosity of Sun)

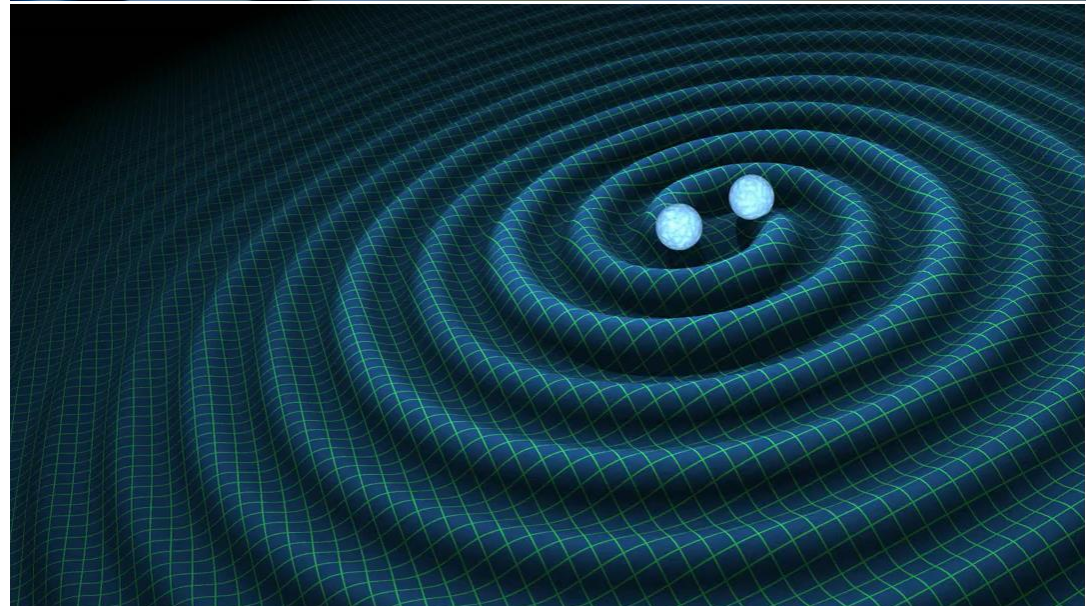


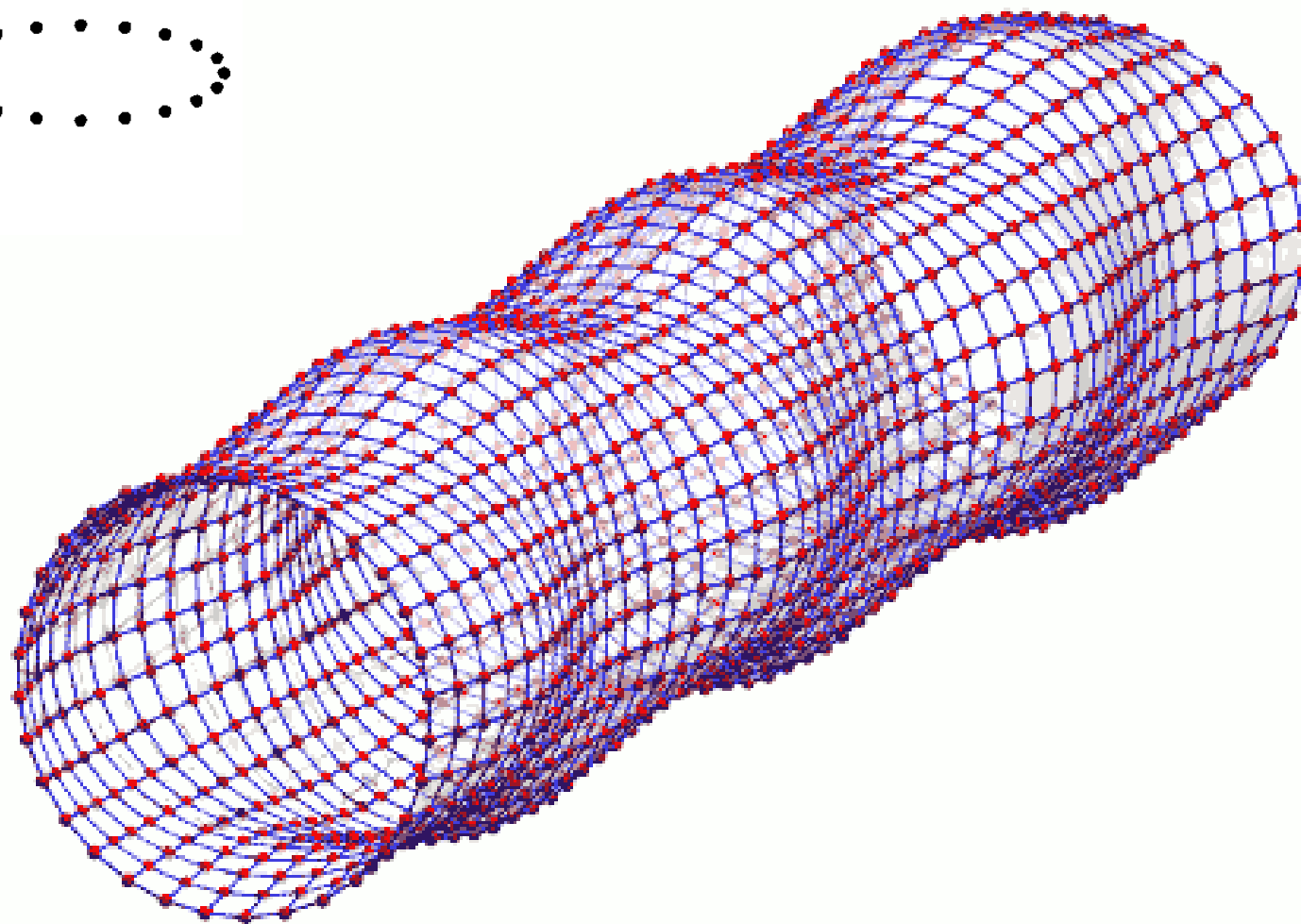
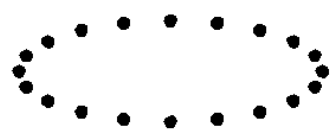
Gravitational waves: Ripples on space-time pond

- Travel with speed of light
- Transverse in nature.
- Radiates away the bulk asymmetry.
- Dominant contribution is quadrupolar in nature. Amplitude of GW is

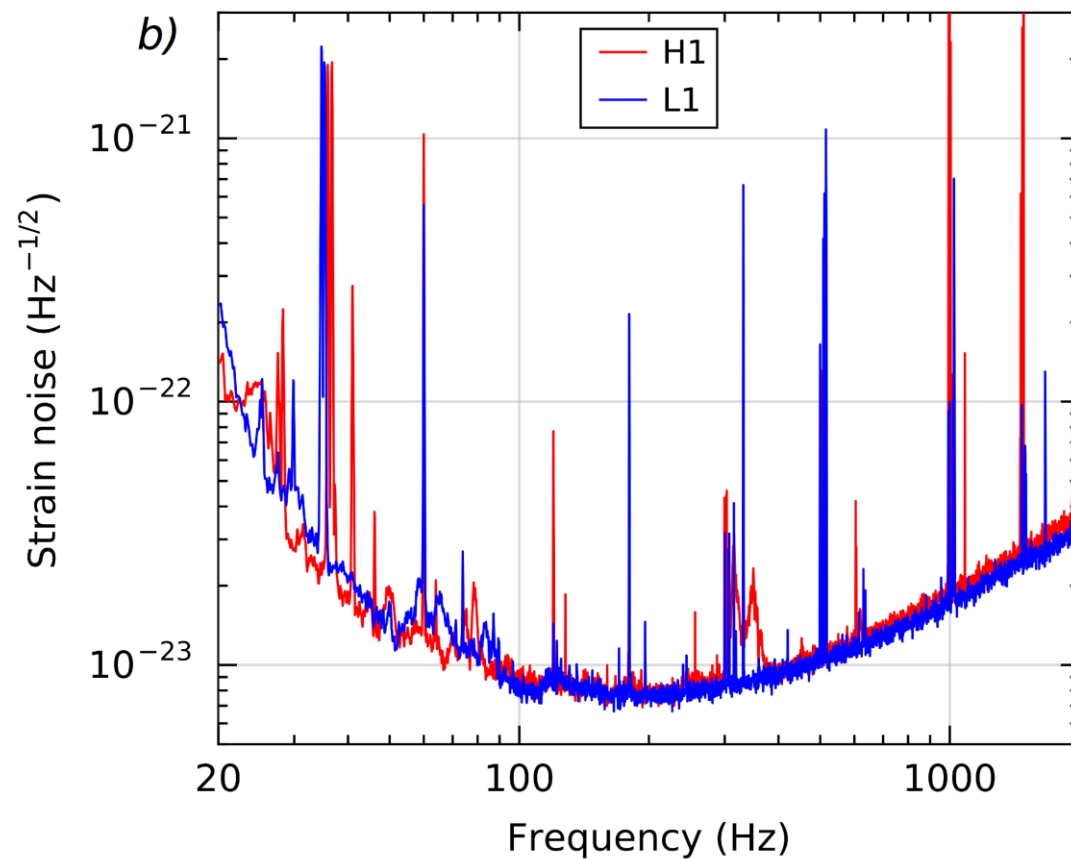
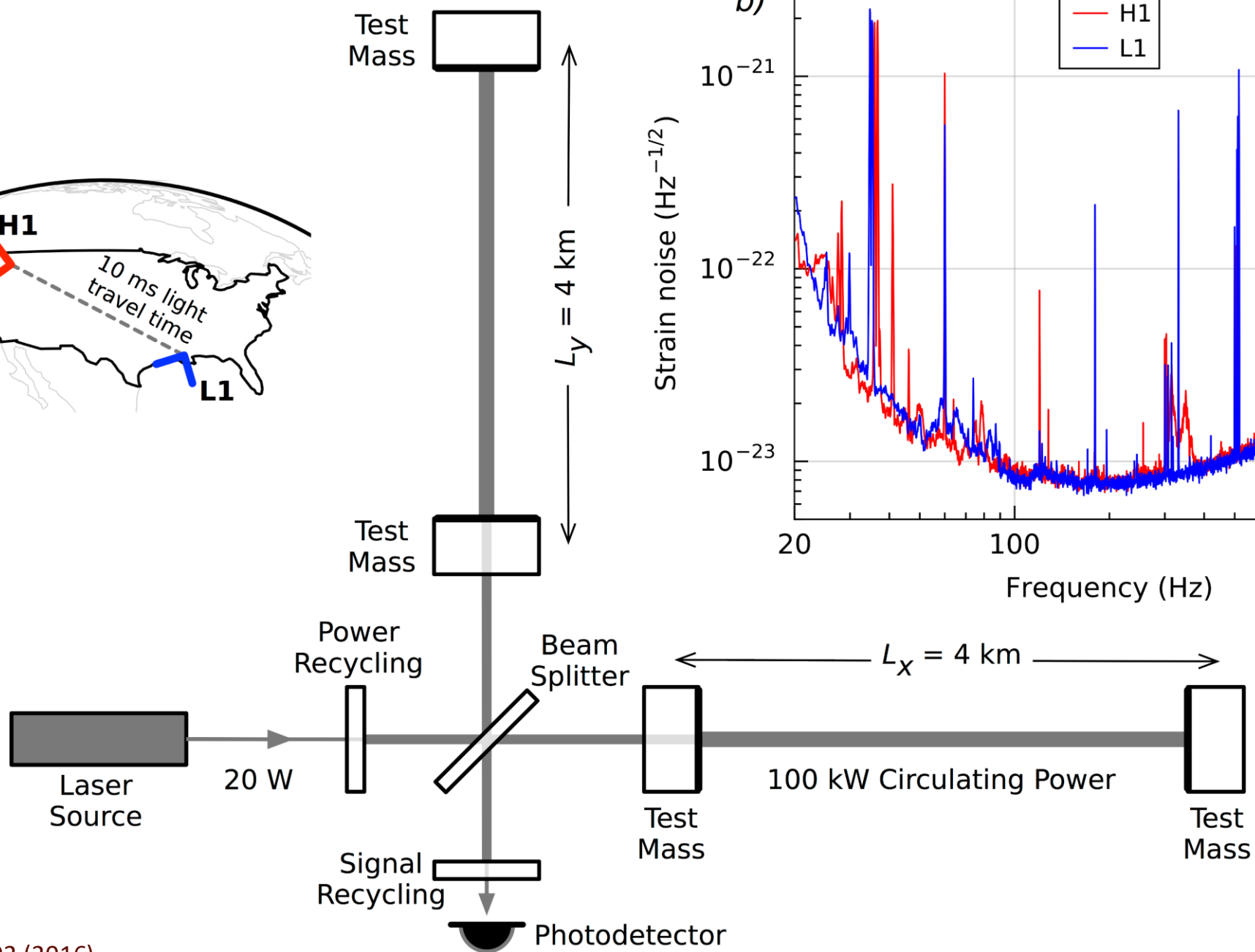
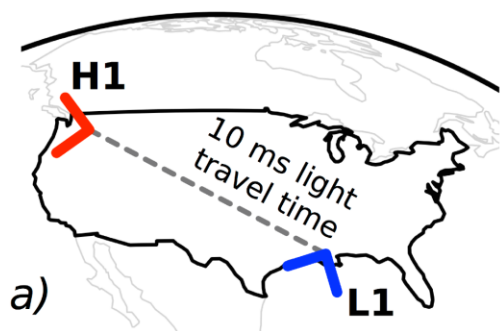
$$h = \frac{2G}{rc^4} \frac{d^2 Q}{dt^2} \quad \text{Q: Quadrupole Moment}$$

$$Q_{ij} = \int \rho(\mathbf{r}) \left(3r_i r_j - \|\vec{r}\|^2 \delta_{ij} \right) d^3 \mathbf{r}$$



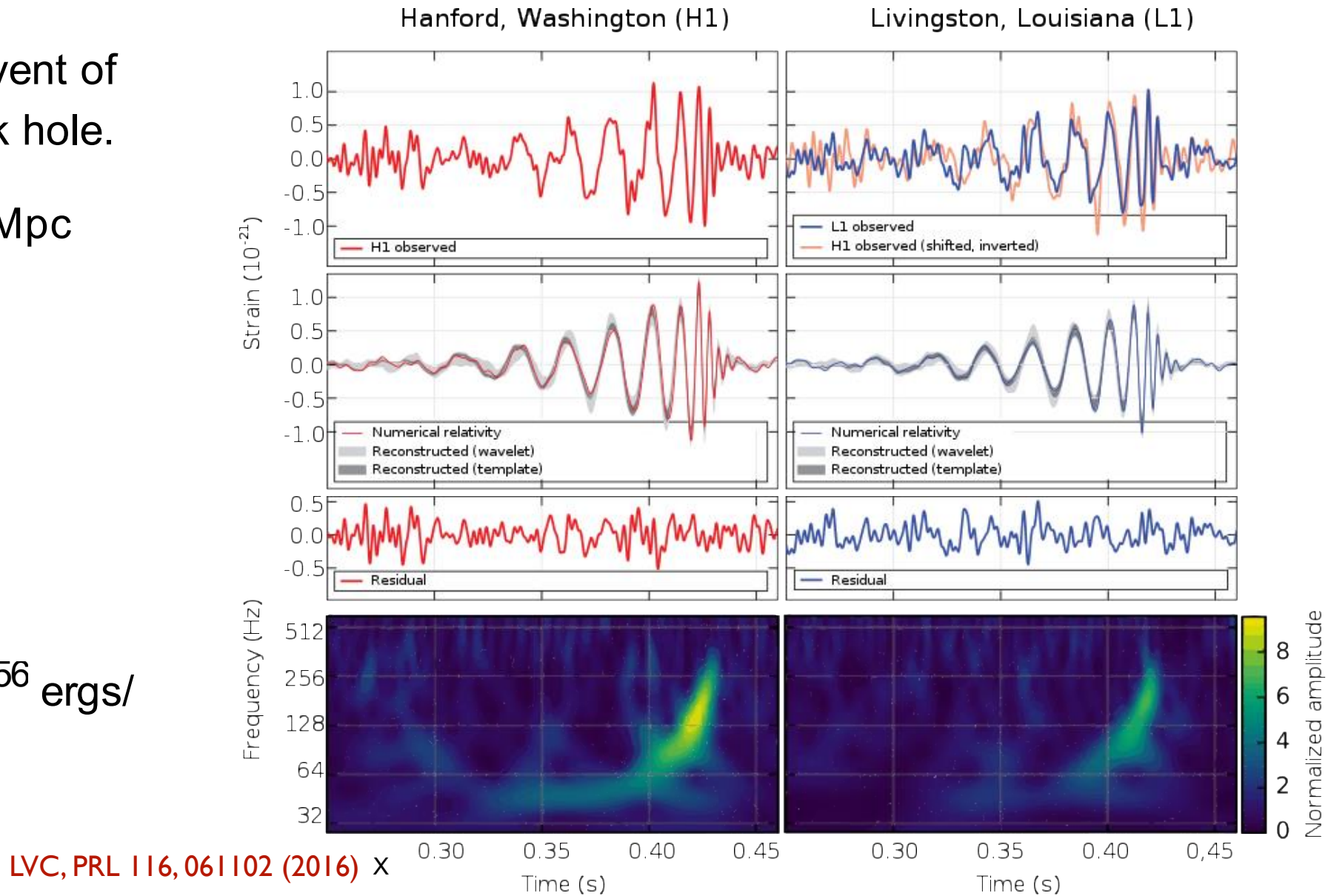


Effect of gravitational waves



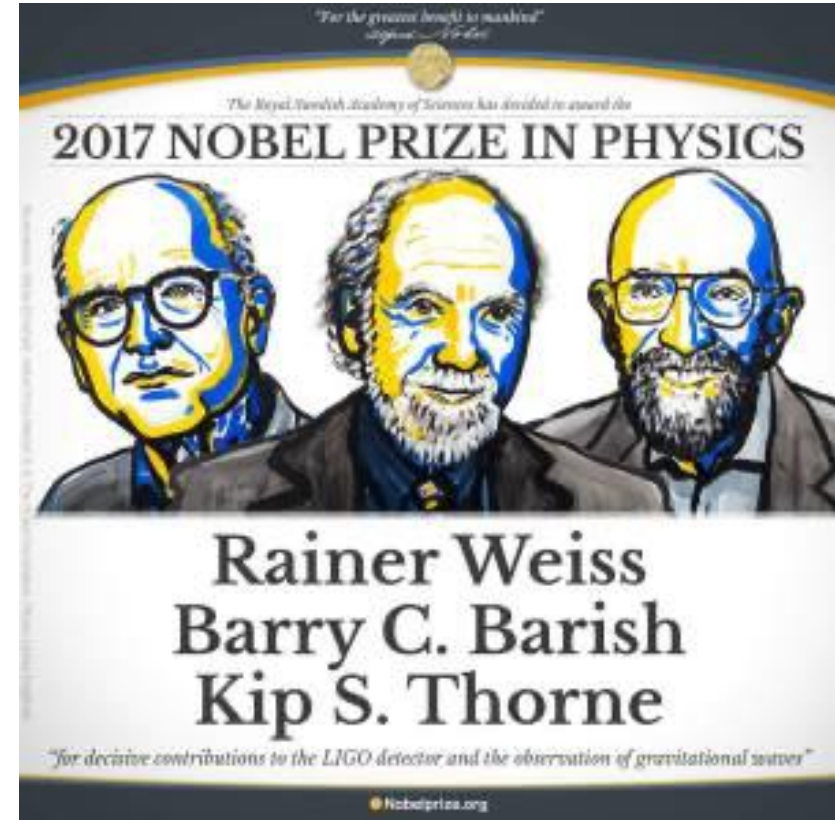
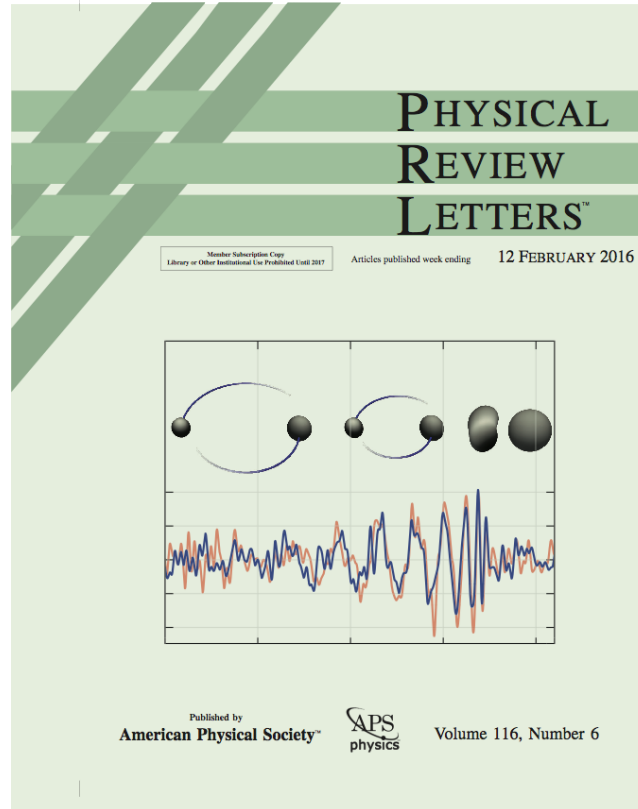
GW150914: First observation of binary BH merger

- Stellar mass BH merger event of $36\text{-}29 M_{\odot}$ into a kerr black hole.
- Luminosity distance: 450 Mpc
- Signal duration : 0.2 sec
- Remnant BH of $62 M_{\odot}$
- Highly relativistic
- Peak strain of $h \sim 10^{-21}$
- Peak Luminosity of $\sim 10^{56}$ ergs/sec
- Sky patch ~ 590 sq. Deg.



2017 Nobel prize in Physics

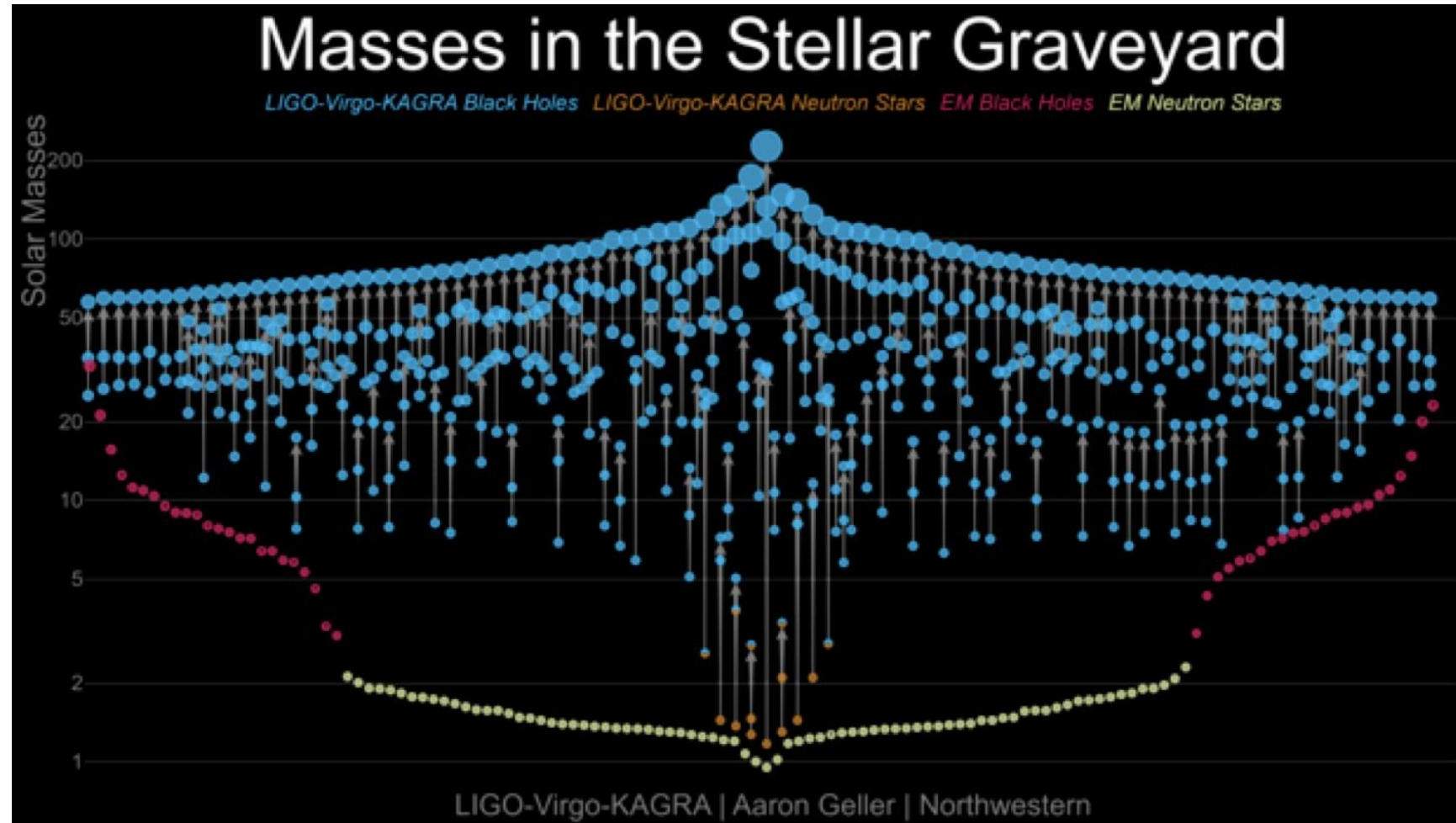
- **First** direct detection of gravitational waves
- **First** direct evidence of collision of black holes in the binary system
- **First** direct observation of stellar mass black holes in a binary system
- **First** direct evidence of existence of stellar mass black holes $> 25 M_{\odot}$



For decisive contributions to the LIGO And
The observation of gravitational waves
2017 Nobel Prize in Physics

Open questions:

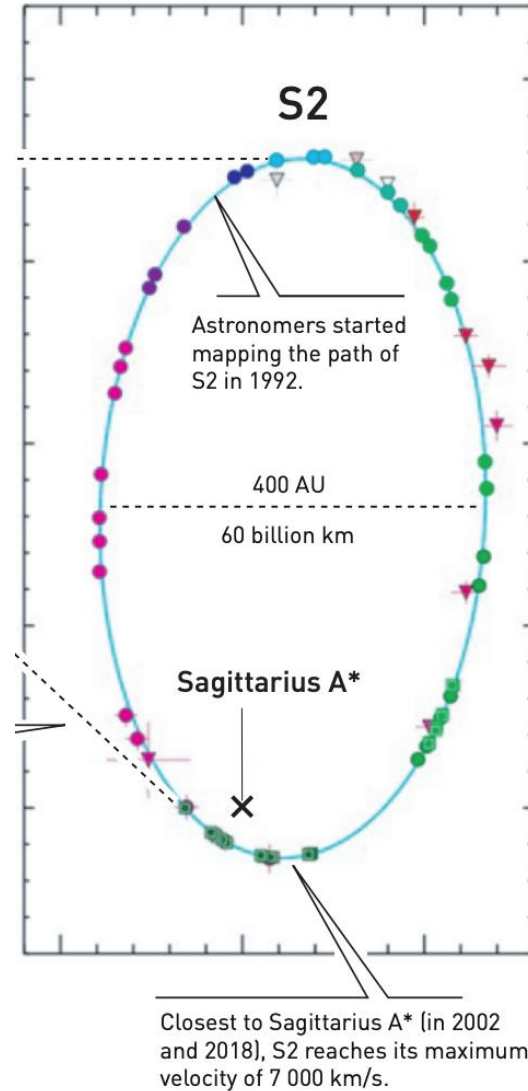
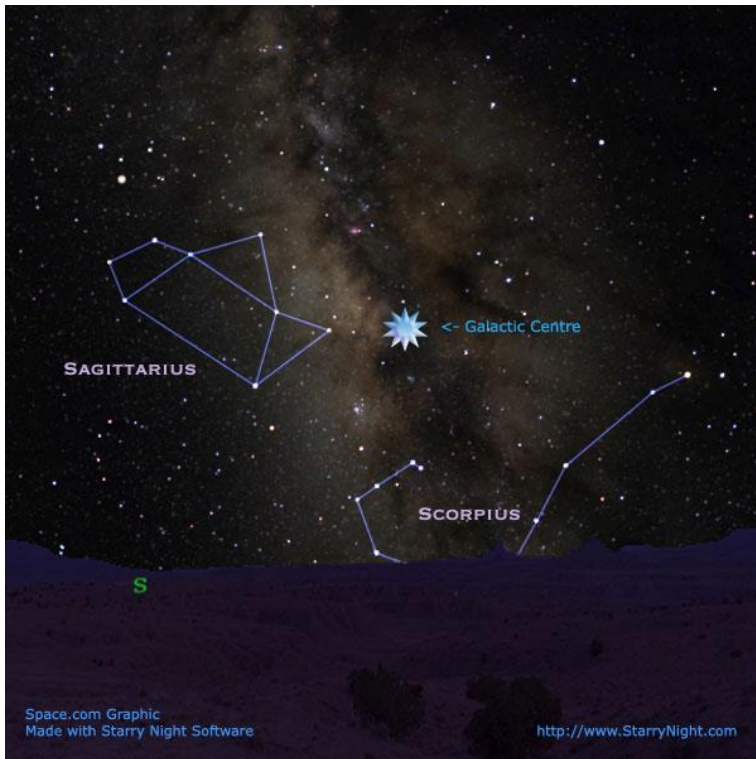
- BH population and distribution in our Universe?
- Formation of intermediate black hole channels?
- SMBH formation and Galaxy formation?
- Testing General Theory of Relativity.



Credit: LVK

Supermassive black holes

- Black holes with million to billion times M_{sun}
- Most of the galaxies harbour supermassive black hole at the centre



© Nobel Media. Ill. Niklas Elmehed.
Reinhard Genzel
Prize share: 1/4



© Nobel Media. Ill. Niklas Elmehed.
Andrea Ghez
Prize share: 1/4

For the discovery of a supermassive compact object at the centre of our galaxy
2020 Nobel Prize in Physics

Map the orbit of S2 using ESO for 16 years. Estimated the mass of SMBH at the centre to be 4 million M_{sun} .

Gamma Ray Bursts

Fermi/GBM satellite, SWIFT (NSAS-GSFC)

Frequent (one per day)

Highly energetic sources with 10^{49} - 10^{53} ergs in few secs.

Ultra relativistic and highly beamed

Two types:

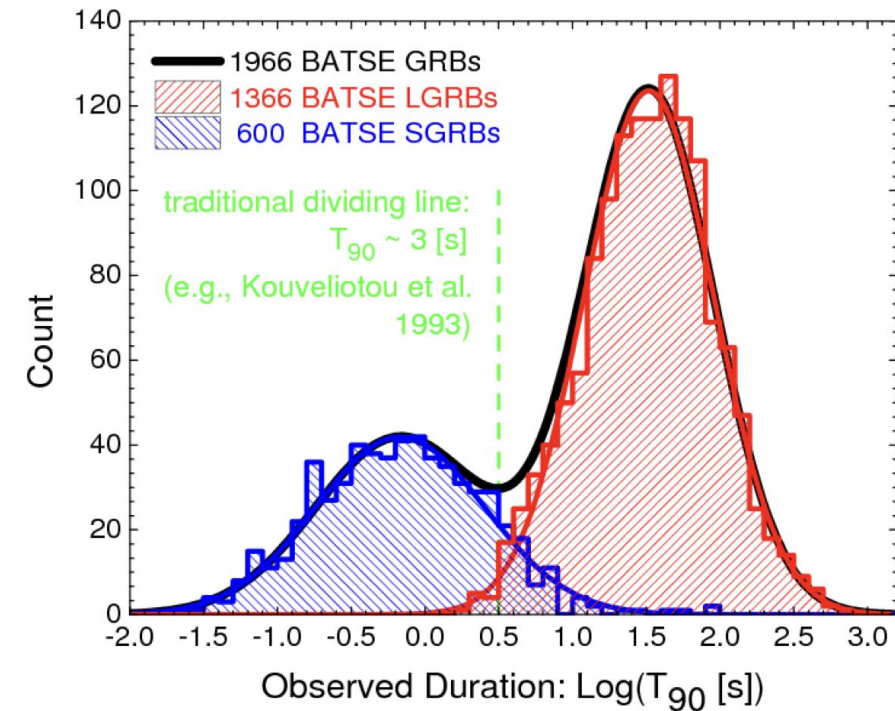
Long $T > 2$ secs and short $T < 2$ sec.

Long GRBs:

Supernova core collapse from the galaxy association

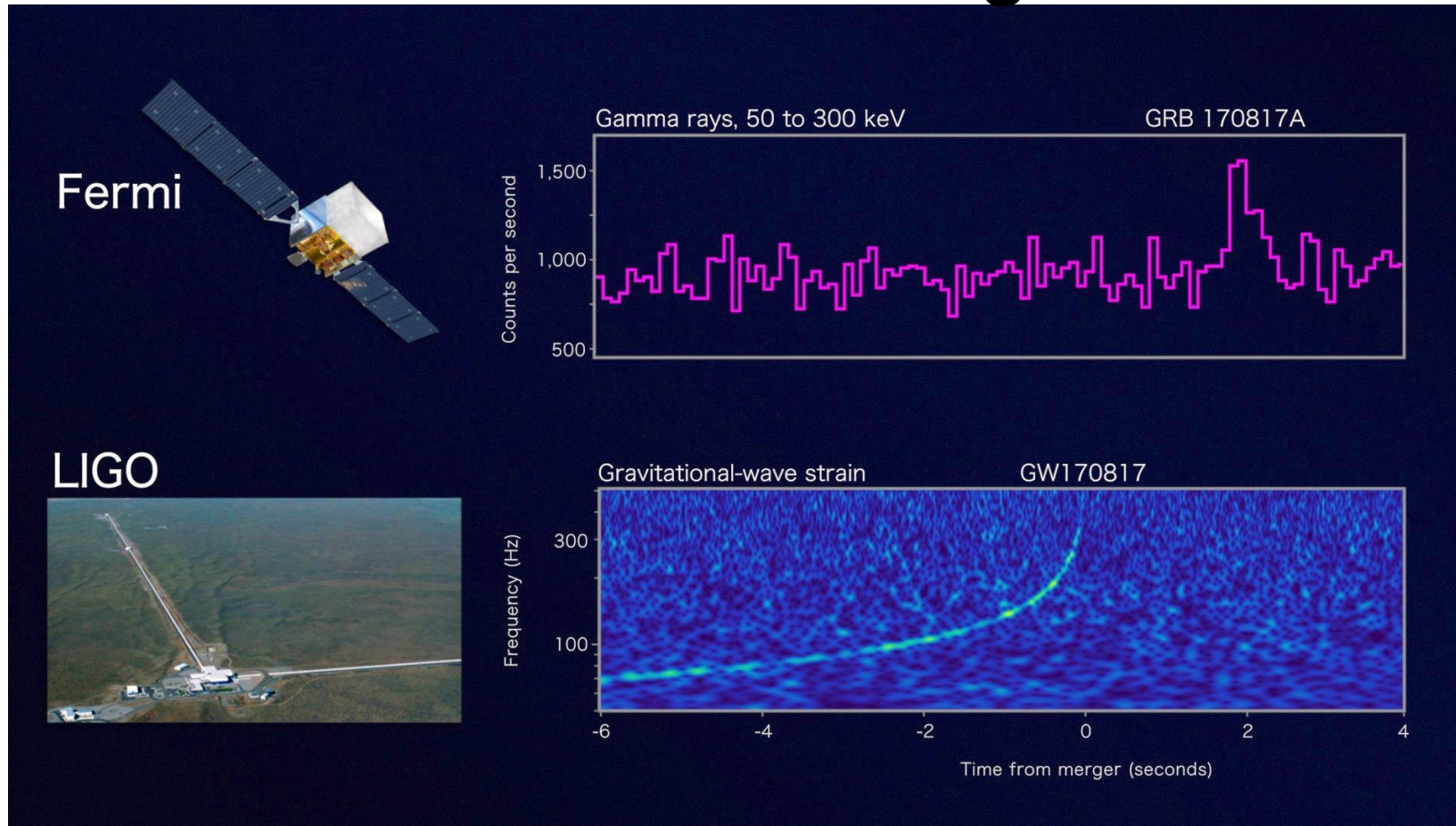
Short GRB:

Binary Newton star merger?? (Puzzle settled few years ago)



Spectacular multi messenger discovery

GW170817: the first binary neutron star merger



- Jointly observed as binary NS merger event in GW window and short GRB in the EM telescopes
- Closest LIGO-Virgo compact binary merger event
- Closest observed short GRB event @40 Mpc

Cosmology model

- Olber's Paradox: Why is the night sky dark?
- If the Universe is **Infinite and Static**, the sky should be uniformly bright.
- What is the Universe is NOT
 - **Static but expanding**: Less light from the distant stars
 - **Infinite but Finite**: Distant light has not reached us.
- Observation of Galaxies and Hubble's Law



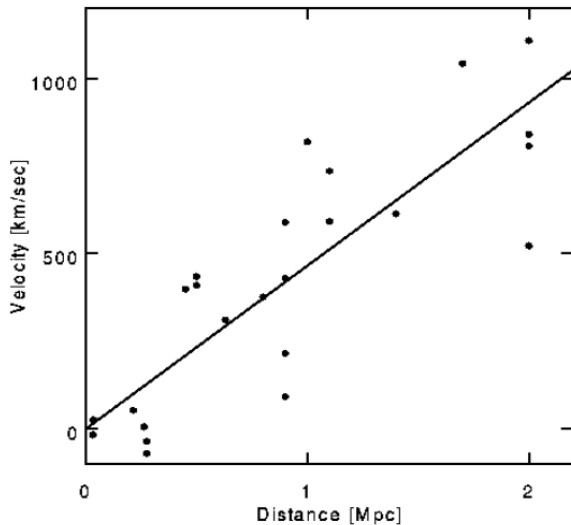
Credit: ESO

Hubble's law and expansion

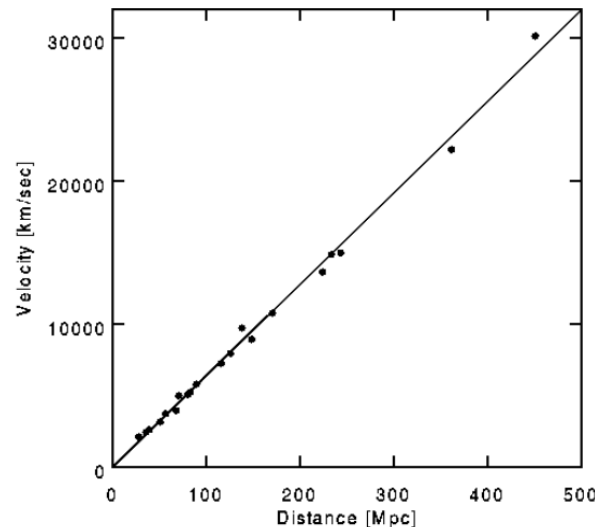
- Distant galaxies showed shift in the frequency.
- Inferred velocity indicated that they are going away --- Expansion!

$$\frac{dR}{dt} \equiv V = H_0 R$$

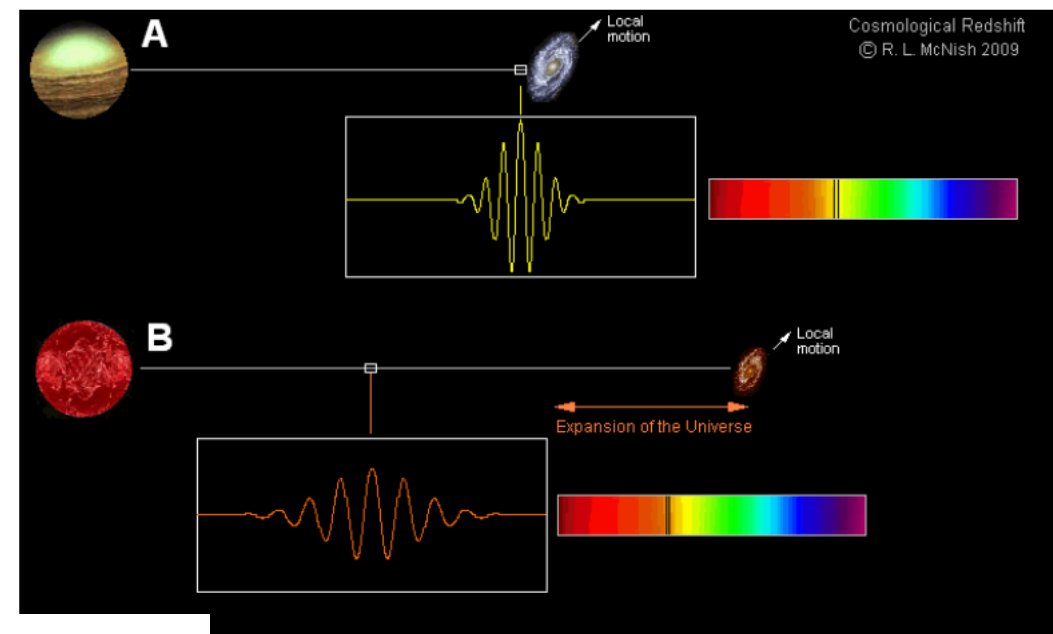
V is velocity of galaxy (km/sec) H_0 is Hubble constant
 R is distance (Mpc) $1pc = 3.26$ light years



Hubble (1929)



Pisces et al (1999)

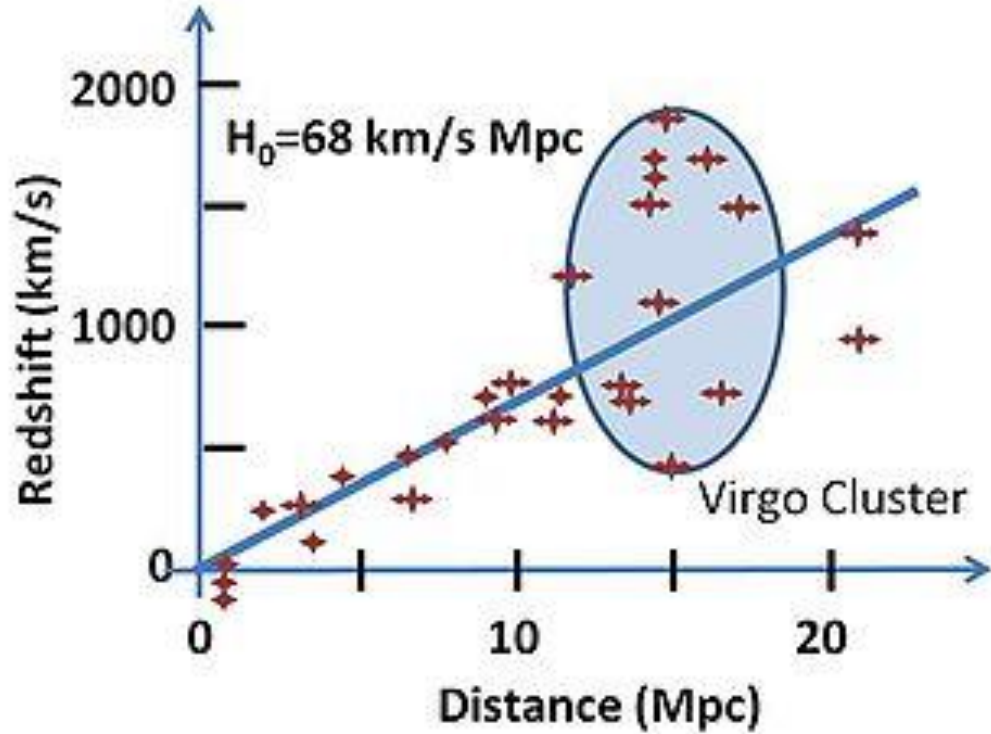


Two galaxies separated by 1 Gpc

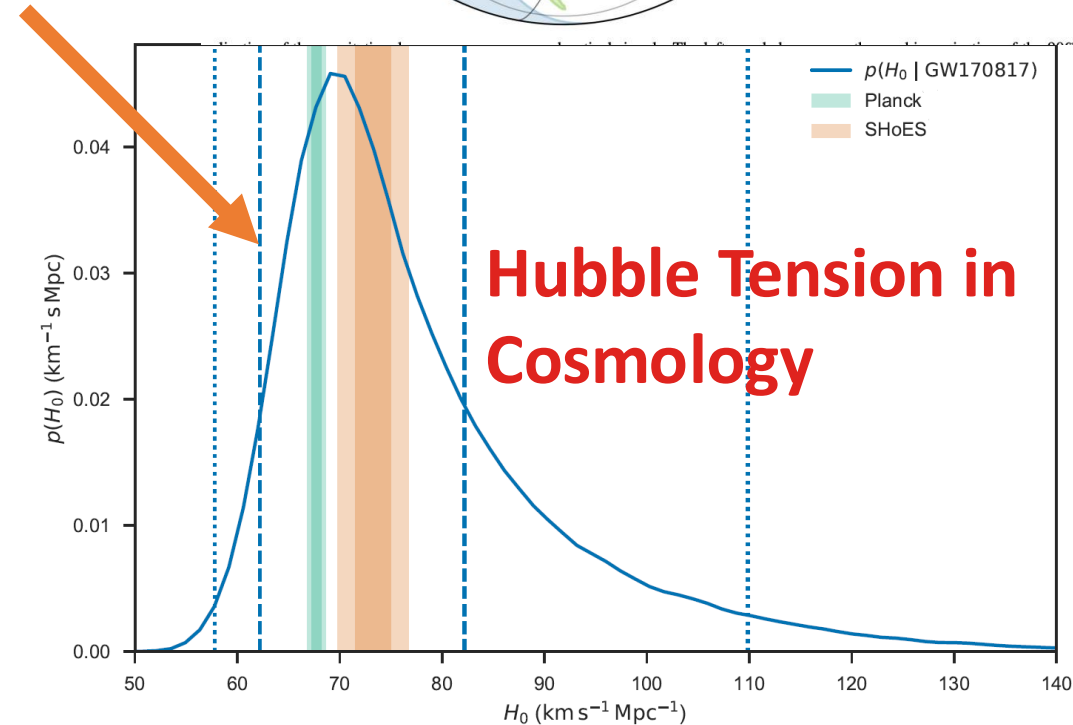
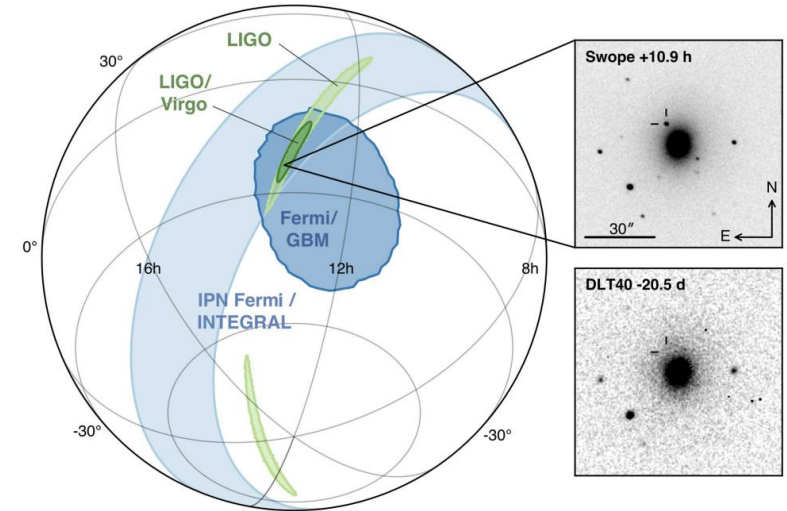
$$v = (70 \text{ km/s/Mpc})(1 \times 10^3 \text{ Mpc}) = 7 \times 10^7 \text{ m/s} \sim 0.2 c$$

Newtonian gravity is not suited!

Cosmology with GW170817



Hubble's Law
 $Velocity = H_0 \times d$
 Hubble's constant



*Optical counterpart from Swope NGC4993 —
 redshift, distance
 Include in the Bayesian inference to obtain
 Hubble's constant
 ~70 km/sec/Mpc*

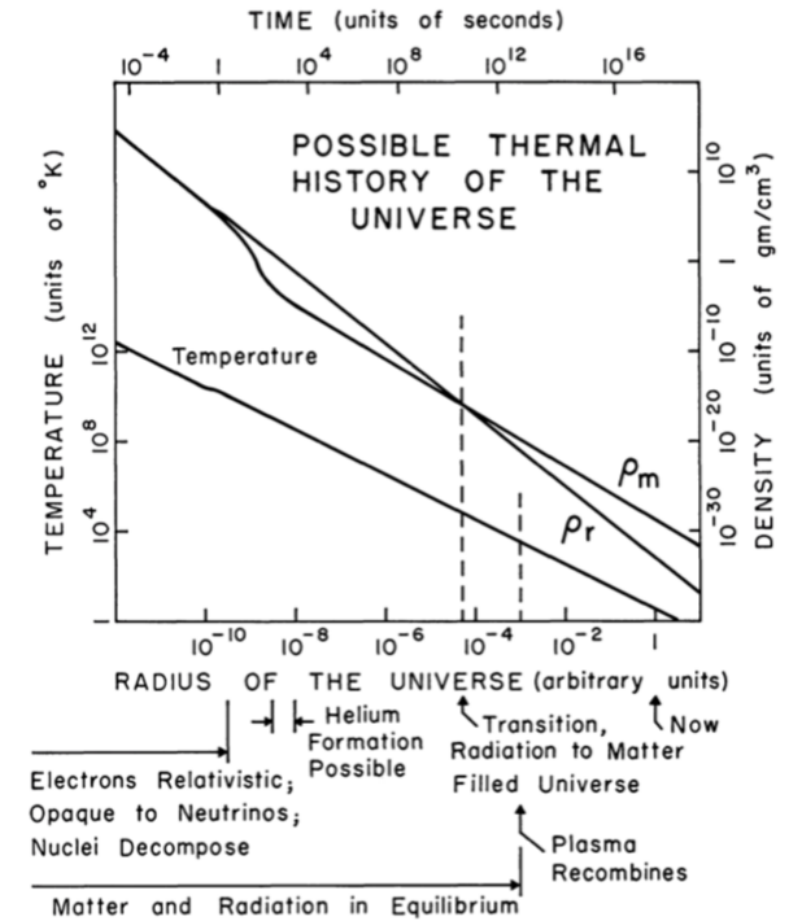
LVC Nature, 2017

MM paper ApJ Lett, 2017

Big Bang and Expansion of the Universe

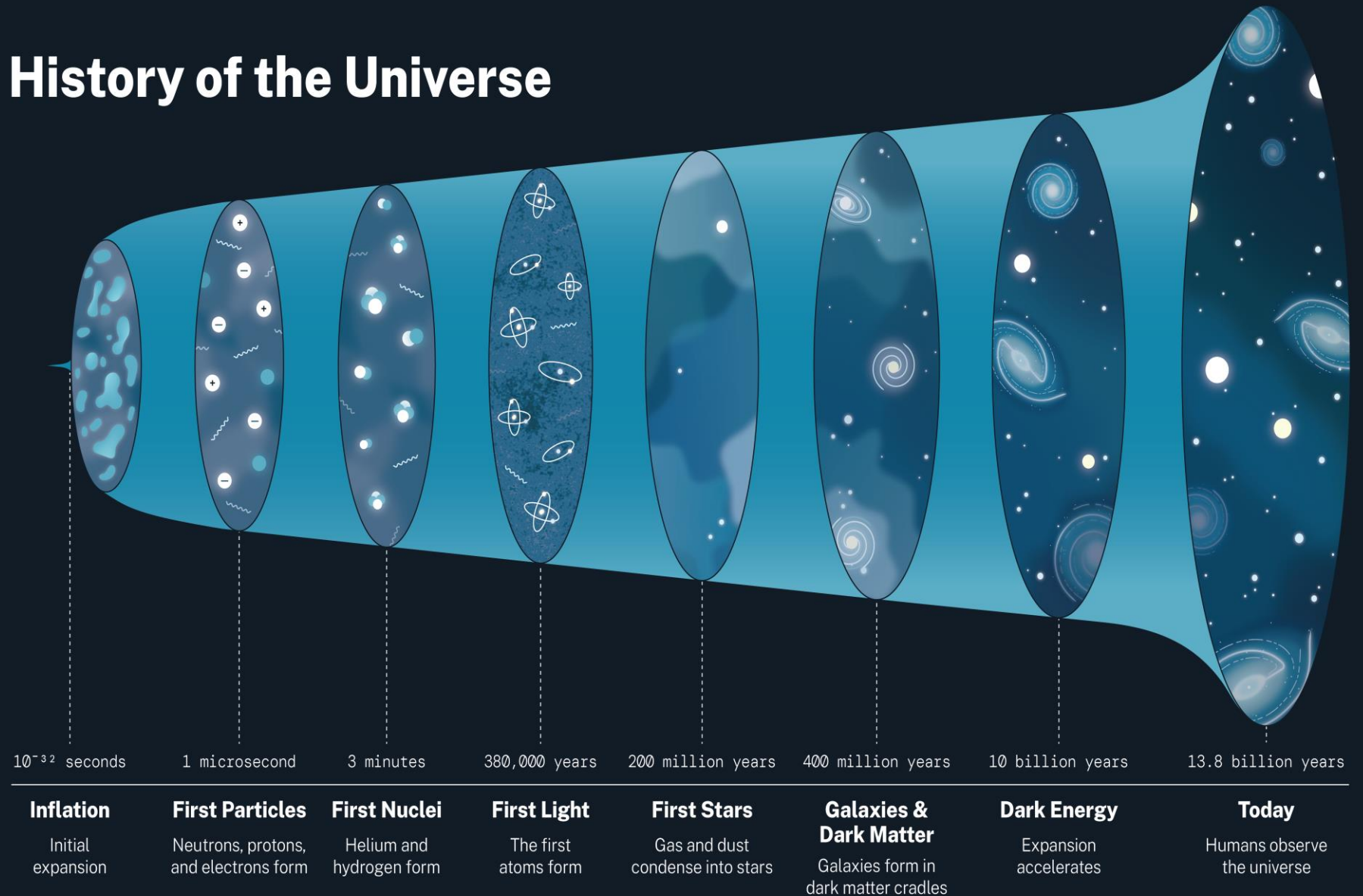
Peebles, Nobel Prize in Physics, 2019

- Friedman-Lemaitre-Robertson-Walker model
 - Expansion of the Universe (captured in the expansion parameter)
 - Homogeneous: Uniform everywhere
 - Isotropic: No preferred direction
- Universe was hot in the beginning and cooled as it expanded
 - Hot, ionized initial state produce thermal radiation
 - Photons decouple when Universe stops being ionized (last scattering)
 - Expansion by $a(t)$ cools black body spectrum from T to $T/a(t)$.
 - An ambient Cosmic Microwave Background (CMB) of photons with Black body spectrum



Thermal History of the Universe

History of the Universe



Observational evidence: COBE

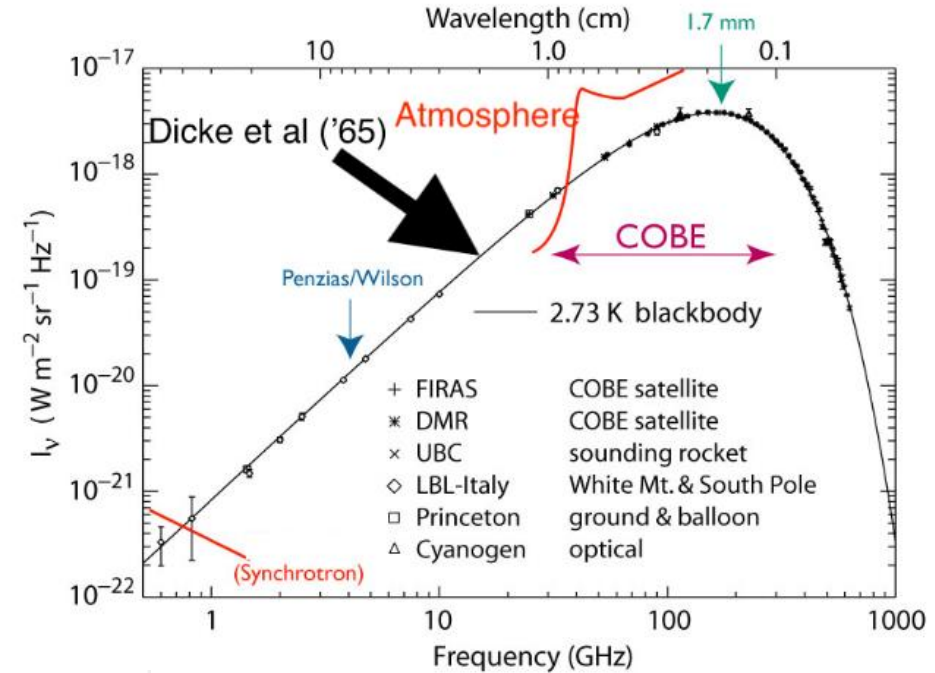
- An ambient **Cosmic Microwave Background** (CMB) of photons with **Black body spectrum**

Penzias & Wilson, Nobel Prize, 1978

- Penzias & Wilson (1964) observed excess **antenna temperature** of **$3.5 \pm 1.0 K$** .

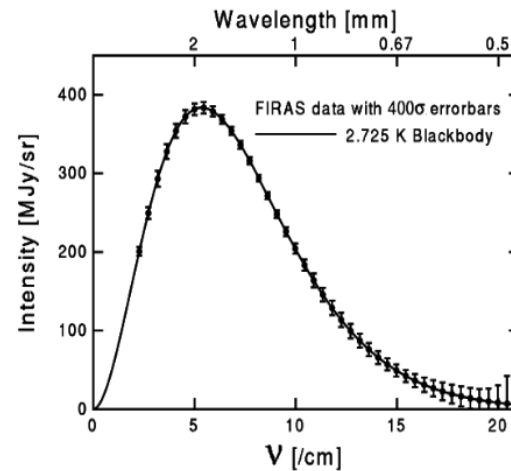
- Dicke et al (1965) could explain this excess using FLRW model.

- COBE measurements (1990) confirm the black-body spectrum.



First evidence of Big-bang

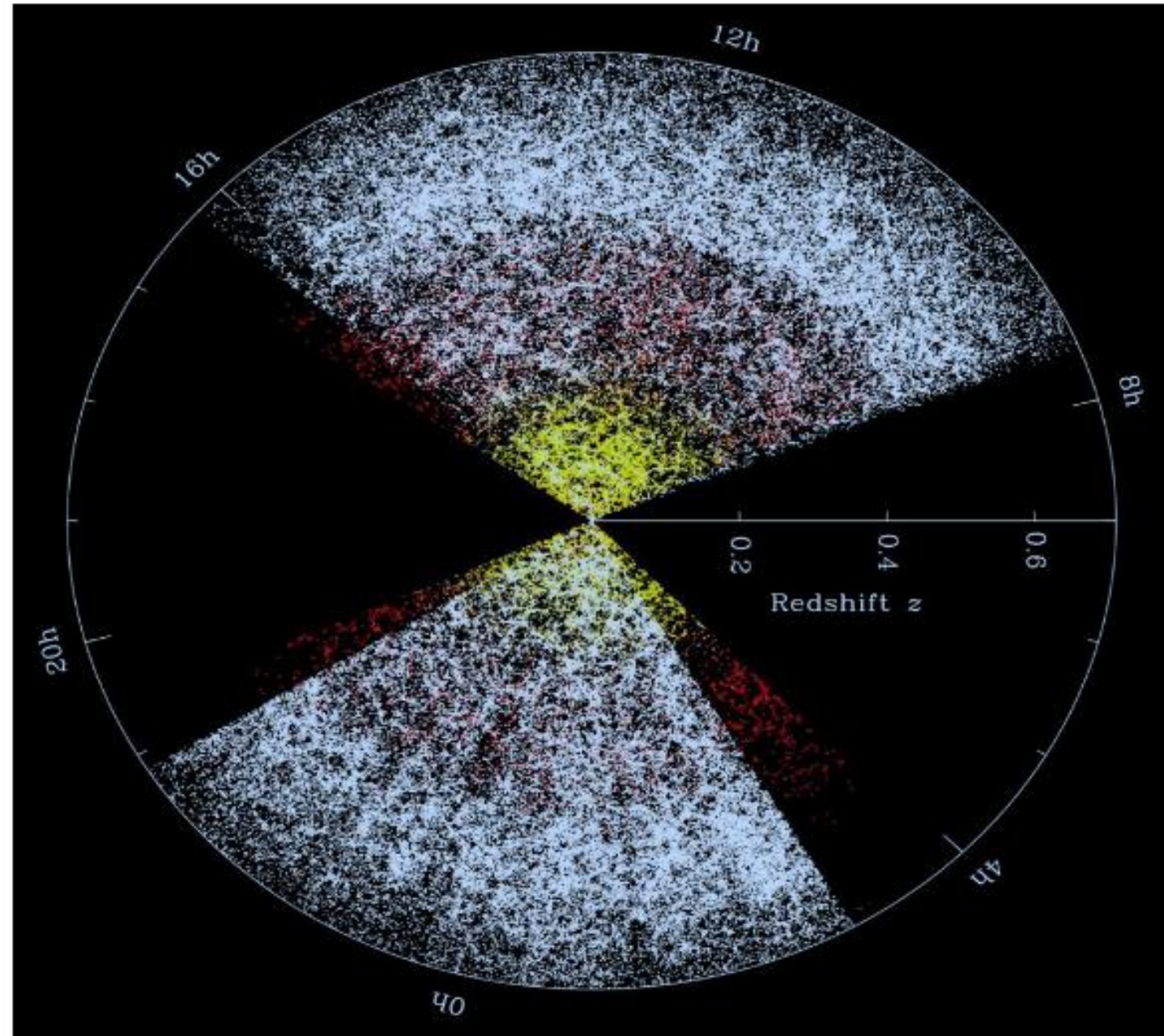
Mather & Smooth, Nobel Prize, 2006



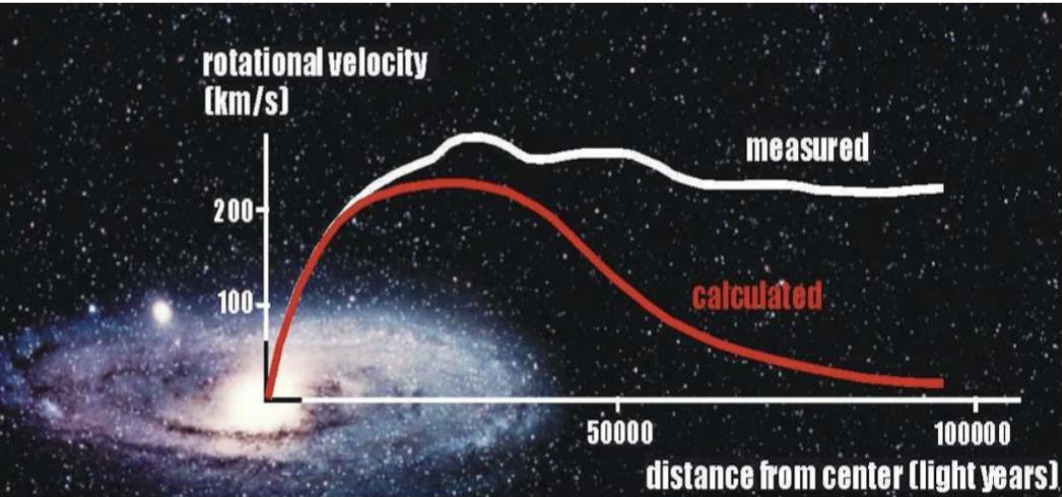
Perfect Black body spectrum from COBE

SDSS Survey (Sloan Digital Sky Survey)

- Multi-spectral imaging spectroscopic digital survey
- Each pixel correspond to a galaxy. Millions of galaxies are mapped
- Confirms homogeneity and isotropy



Evidence of Dark Matter



$$v^2(r) = \frac{GM(r)}{r}$$

$M(r)$ is the total mass within radius r

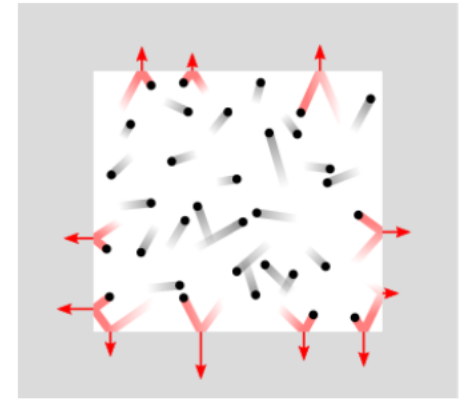
- If the galaxy mass is restricted to the visible mass, the rotational velocity should decrease based on Kepler's law.
- Evidence of existence of Dark Matter : Has mass and interacts only via gravity
- Galaxy requires extended DM halos



- Dark Matter: Proposed by Zwicky
- Vera Rubin (Rubin and Ford, 1970)
 - Observed discrepancy in the angular motion predicted using visible light and actual angular motion
 - The outer portions of the galaxy were rotating at much higher velocity
 - Something is holding them together

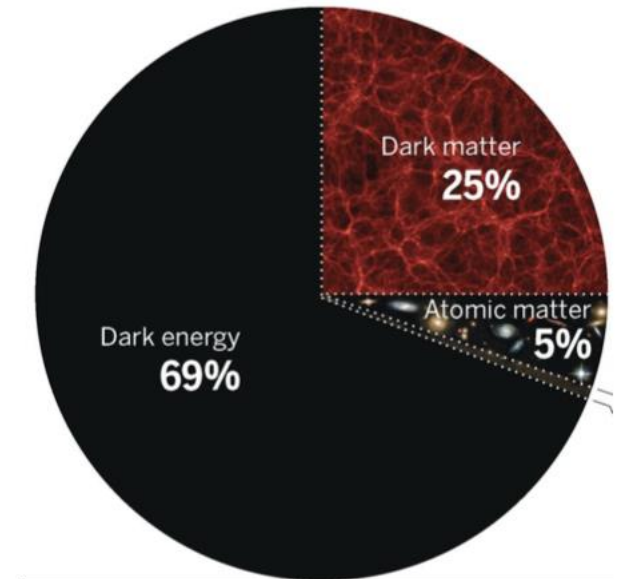
Dark Energy

- Pressure is the collisions by the molecules on the container.
- Increase the number of gas molecules, increases the number of collisions and hence, pressure increases.
- For Dark Energy, increase in the 'molecules', reduces the pressure. It depends on the 'Empty' space
- As the Universe expands, the Dark Energy increases and accelerates the Universe further.



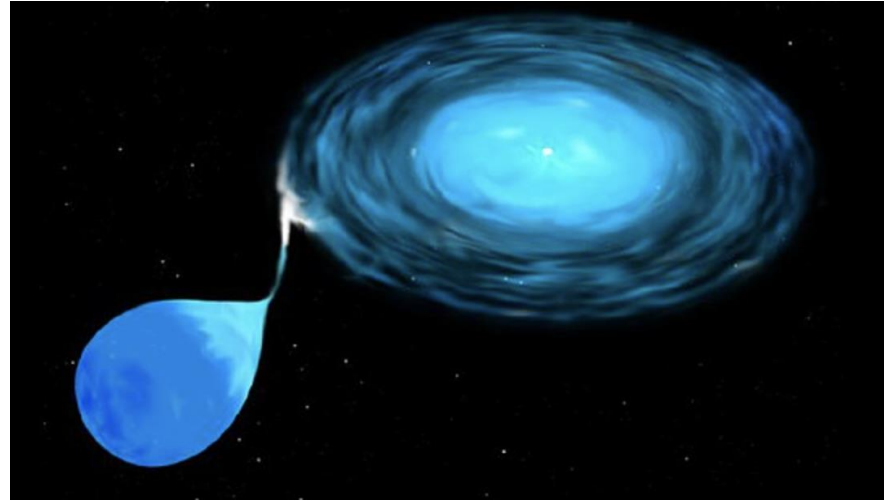
Credit: Wikipedia

Credit: Spergel ('15)

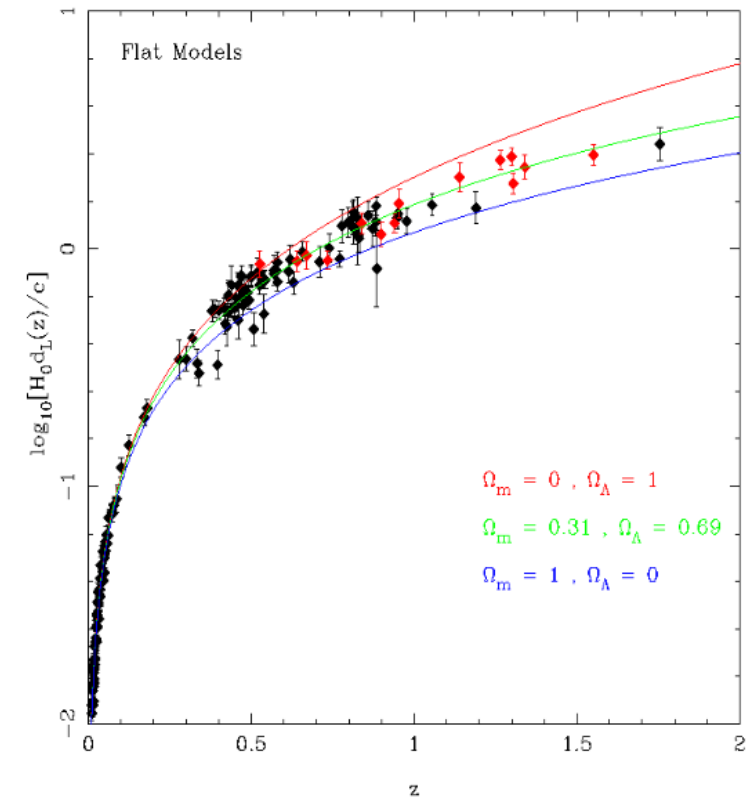
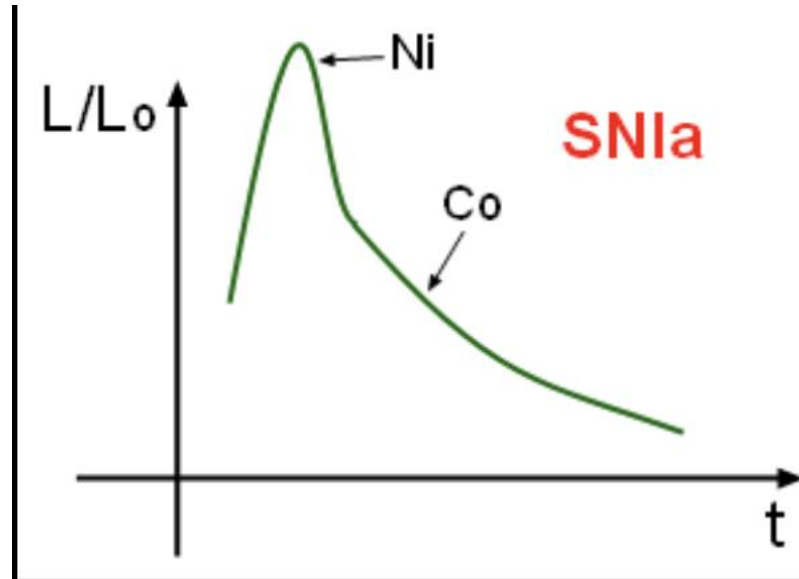


Accelerated Expansion of the Universe

Accreting White-Dwarf



- Nobel Prize in Physics (2011): Perlmutter+Schmidt+Reiss
- for the discovery of the accelerating expansion of the Universe through observations of distant supernovae.
- Observed ~ 50 SN-IA which act like a standard candle with a characteristic light curve
- SN appeared to be much more fainter than expected.



H_0 tension

Credit: Cosmoverse arXiv:2504.01669

Local measurements

Based on distant SNe-Ia using background Hubble flow, calibrated using Cepheids etc:

$$H_0 \sim 72 - 74 \text{ km/s/Mpc}$$

Cosmic measurements

Based on modeling measurements of the fluctuations in CMB under the assumptions of Λ CDM:

$$H_0 \sim 67 \text{ km/s/Mpc}$$

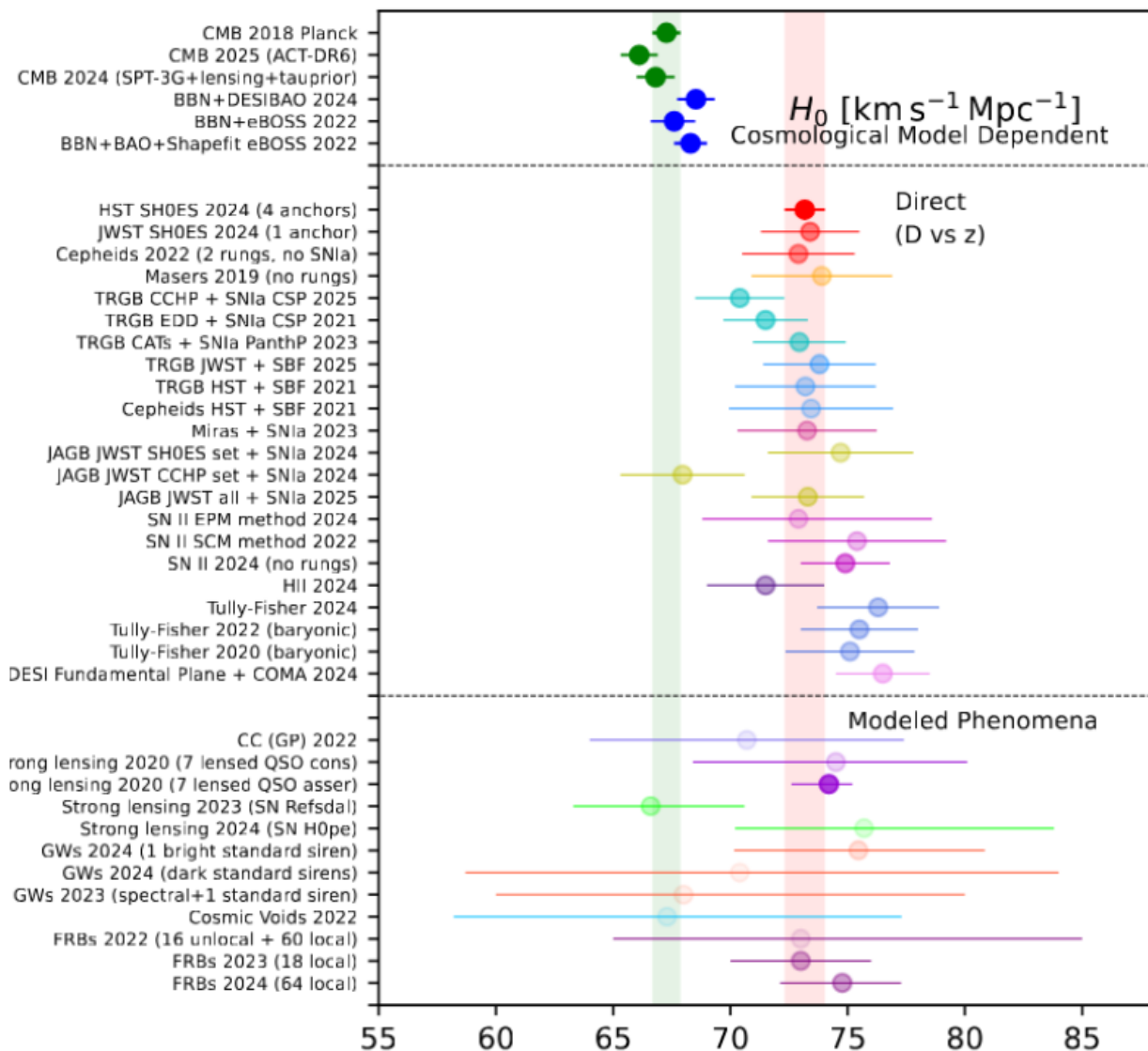


Image credit: NASA



1990

Ground-based observatories



1995

Hubble Deep Field



2004

Hubble Ultra Deep Field



2010

Hubble Ultra Deep Field-IR



FUTURE

James Webb Space Telescope



Redshift (z):

Time after
the Big Bang

Present

1

6
billion
years

4

1.5
billion
years

5

6

7

800
million
years

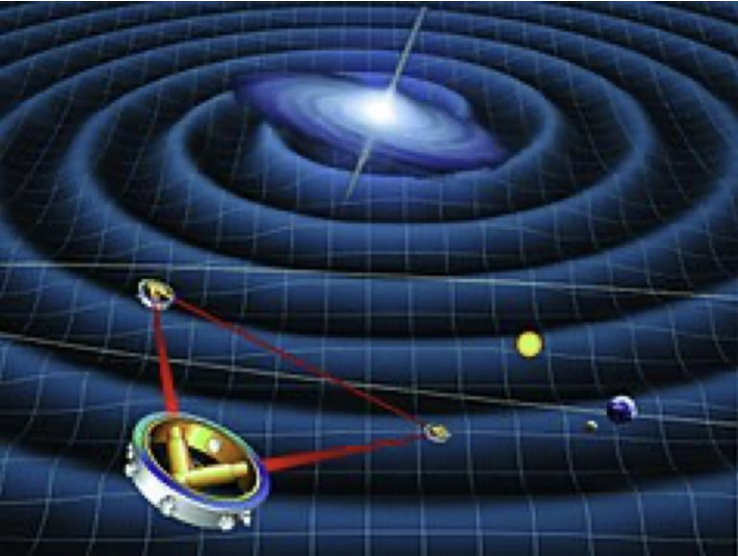
8

10

480
million
years

>20

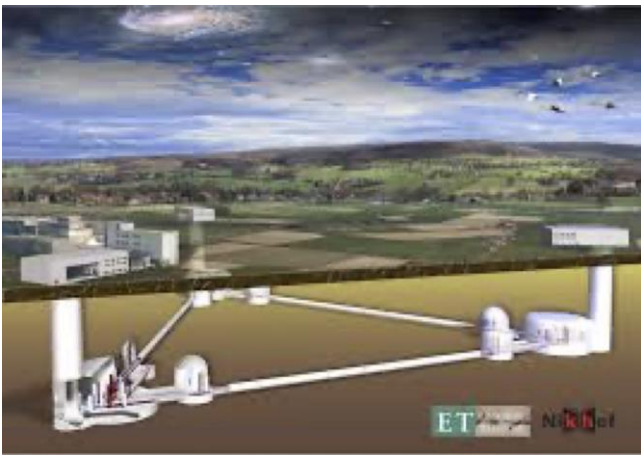
200
million
years



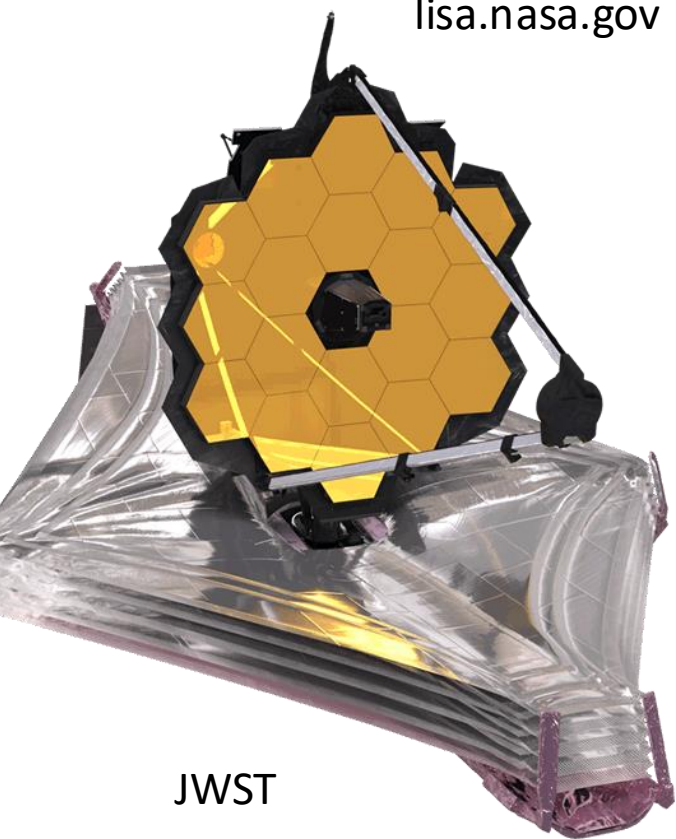
lisa.nasa.gov



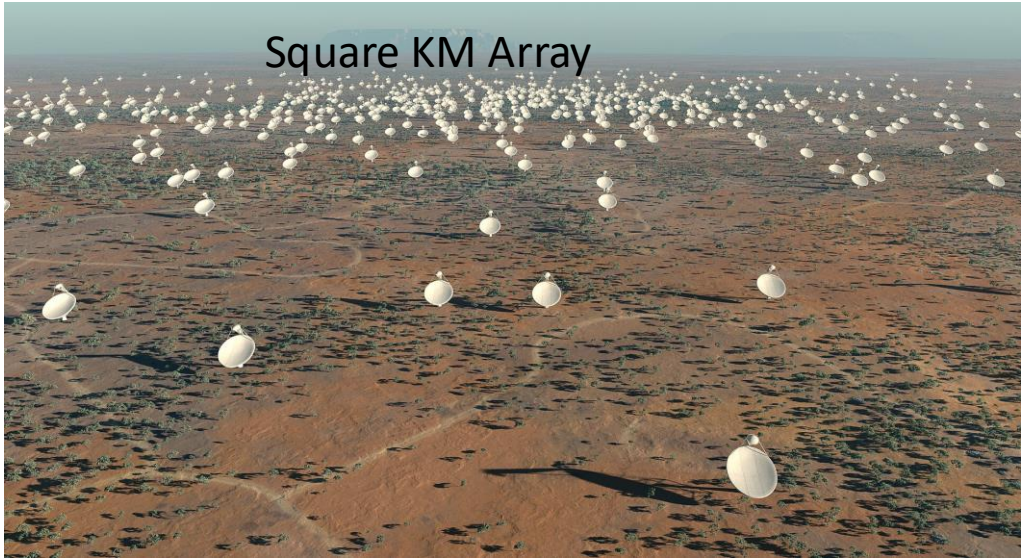
VeraRubin



Einstein Telescope : Third generation underground detector (Puntoro etal CQG2010)



JWST



Square KM Array